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10. LANDSCAPE AND VISUAL IMPACT ASSESSMENT

10.1 Introduction

10.1.1 The assessment of potential landscape and visual impacts due to construction and operation of the Project is presented in this chapter.

10.1.2 Landscape and visual impacts were assessed in accordance with the criteria and guidelines as stated in Annexes 10 and 18 of the Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM) and the latest version of Environmental Impact Assessment ordinance (EIAO) Guidance Note No. 8/2023 on "Preparation of Landscape and Visual Impact Assessment under the Environmental Impact Assessment Ordinance".

10.2 Scope and Content of the Study

Assessment Area

10.2.1 The assessment area for landscape impact assessment includes all areas within a 100m from the Project boundary in accordance with EIA Study Brief (No. ESB-375/2025). The assessment area for the visual impact assessment is defined by the visual envelope of the Project. The landscape impact study boundary are shown in Figures **60722948/A10/FIGURE 10.1** to **10.7**. The aerial photos overlaid with the project layout are shown in Figures **60722948/A10/FIGURE 10.8** to **10.14**.

Landscape Impact Assessment

10.2.2 The existing and planned landscape resources and character within the assessment area were described, appraised, analysed and evaluated. A system is derived for judging landscape and visual impact significance as required under the Technical Memorandum. The sensitivity of the landscape framework and its ability to accommodate change are particularly focused on. The degree of compatibility of the Project with the existing and planned landscape setting was identified. The landscape impact assessment evaluated the potential landscape impact so as to illustrate the significance of such impacts arising from the Project. Clear mapping of the baseline landscape resources, landscape character areas and the landscape impact were provided.

Visual Impact Assessment

10.2.3 The visual impact of the Project was assessed. For aboveground structures of the Project, clear illustrations including mapping of visual impact are provided. The assessment adopts a systematic methodology and includes the following:-

- Identification and plotting of visual envelope of the Project;
- Appraisal of existing visual resources and character as well as the future outlook of the visual system of the assessment area;
- Identification and justification of the key public viewing points within the visual envelope and their views at sea level, ground level and elevated vantage points, and clearly indicate the key public viewing points on a plan of appropriate scale;

- Evaluation of the magnitude of change in terms of visual composition, visual obstruction and visual change of the Project with the existing and planned visual context, and sensitivity of viewers in terms of types of viewers and value of existing views;

- The visual impact of the Project with and without mitigation measures during operational phase shall be included and illustrated so as to demonstrate the effectiveness of the proposed mitigation measures across time; and

- Evaluation and explanation with supportive arguments of factors considered in arriving the significant thresholds of visual impacts. The visual impacts included the presentation of an evaluation matrix derived for judging impact significance.

Landscape and Visual Mitigation Measures

10.2.4 The merit of preservation in total or in parts of existing landscape and the establishment of a new landscape character are evaluated. Alternative construction methods and / or Project-related works or structure(s) that would avoid or reduce the identified landscape and visual impacts are considered and evaluated for comparison before adopting other mitigation or compensatory measures to alleviate the impacts. The mitigation measures proposed are not only concerned with damage reduction but also included consideration of potential enhancement of the existing landscape and visual quality. Mitigation measures to minimise the adverse effects identified, including provision of a landscape design, are recommended.

10.2.5 The mitigation measures such as tree preservation and tree transplanting as far as practical, provision of buffer planting, reinstatement of affected landscape area, compensatory planting, erection of decorative screen hoarding that compatible with the surrounding setting, reprovision of amenity areas / open spaces, aesthetic design and layout of structures and façade treatment etc. Parties are identified for the on-going management and maintenance of the proposed mitigation works to ensure their effectiveness throughout the operation of the Project. A practical programme and funding proposal for the implementation of the recommended measures are provided.

Significance of Landscape and Visual Impact

10.2.6 Annotated illustration such as coloured perspective drawings, plans and section/elevation diagrams, oblique aerial photographs, photographs taken at vantage points and computer-generated photomontage are adopted where appropriate to illustrate the significance of the landscape and visual impacts of the aboveground ancillary structures of the Project.

10.3 Environmental Legislation, Standards and Guidelines

10.3.1 The following legislation, standards and guidelines are applicable to landscape and visual impact assessment associated with the construction and operation of the Project:-

- Environmental Impact Assessment Ordinance (EIAO) (Cap.499 S.16) and the Technical Memorandum on EIA Process (EIAO-TM), particularly Annexes 10 and 18;

- Environmental Impact Assessment Ordinance Guidance Note No. 8/2023;

- Town Planning Ordinance (Cap. 131);

- Forests and Countryside Ordinance (Cap. 96) and its subsidiary legislations;
- Plant Varieties Protection Ordinance (Cap. 490);
- Protection of Endangered Species of Animals and Plants Ordinance (Cap. 586);
- Hong Kong Planning Standards and Guidelines (HKPSG) Chapters 4, 10 and 11;
- Charter on External Lighting;
- Guideline on Industry Best Practices for External Lighting Installations;
- AFCD Nature Conservation Practice Note No. 2 - Measurement of Diameter at Breast Height (DBH);
- AFCD Nature Conservation Practice Note No. 3 - The Use of Plant Names;
- DEVB TC(W) No. 4/2020 - Tree Preservation;
- DEVB TC(W) No. 5/2020 - Registration and Preservation of Old and Valuable Trees;
- DEVB TC(W) No. 3/2024 - Allocation of Space for Quality Greening on Roads;
- DEVB TC(W) No. 6/2015 - Maintenance of Vegetation and Hard Landscape Features;
- DEVB TC(W) No. 9/2020 - Blue-Green Drainage Infrastructure;
- LAO PN 6/2023 – Processing of Tree preservation and Removal Proposals for Building Development in Private Projects – Compliance with Tree Preservation Clause under Lease;
- GEO Publication 1/2011 - Technical Guidelines on Landscape Treatment for Slopes;
- GEO Publication (1999) - Use of Vegetation as Surface Protection on Slopes;
- Guidelines on Tree Transplanting (September 2014) issued by Greening, Landscape and Tree Management (GLTM) Section of DevB;
- Guidelines on Tree Preservation during Development (April 2015) issued by GLTM Section of DevB;
- Guidelines for Tree Risk Assessment and Management Arrangement issued by DEVB; and
- Study on Landscape Value Mapping of Hong Kong.

10.4 Assessment Methodology

Landscape Impact Assessment Methodology

10.4.1 The landscape impact was assessed according to the following procedures:-

- ***Carry out Broad Brush Tree and Vegetation Survey on Old and Valuable Trees (OVTs) and Trees of Particular Interest within the Assessment Area***. The survey would be carried out within the 100m from the boundary of the Project, which is in accordance with the EIA Study Brief (No. 375/2025), to identify the dominant tree species, maturity, rarity, and distinctive landscape resources (e.g. OVT and Trees of Particular Interest) etc. for the baseline study on the Landscape Resources (LRs) and Landscape Character Areas (LCAs).

- ***Identification and description of the baseline LRs and LCAs found within the Assessment area***. The assessment area includes all areas within 100m of the Project area. This is achieved by site visits and desktop study of topographical maps, information databases and photographs.

- ***Assessment of the degree of sensitivity of the LRs and LCAs***. This is influenced by a number of factors including whether the resource/character is common or rare, whether it is considered to be of local, regional, national or global importance, whether there are any statutory or regulatory limitations/requirements relating to the resource, the quality of the resource/character, the maturity of the resource and the ability of the resource/character to accommodate change.

- The sensitivity of each LR and LCA is classified as follows:

  High: Important landscape resources and landscape character of particularly distinctive in character or high importance, sensitive to relatively small change.

  Medium: Landscape resources and landscape character of moderately valued landscape characteristics reasonably tolerant to change.

  Low: Landscape resources and landscape character, the nature of which is largely tolerant to change.

- Identification of potential sources of landscape impacts. These are the various elements of the construction works and operation procedures that would generate landscape impact.

- Assessment of the potential magnitude of landscape changes. Factors considered include:-

  - the compatibility with the surrounding landscape;

  - the duration of the impact under construction and operational phases;

  - scale of development; and

  - reversibility of change

- The magnitude of landscape changes is classified as follows:-

Substantial: The landscape resources and landscape character would incur a major change.

Moderate: The landscape resources and landscape character would incur a moderate change.

Slight: The landscape resources and landscape character would incur slight or barely perceptible change.

Negligible: The landscape resources and landscape character would incur no discernible change.

- ***Identification of potential landscape mitigation measures***. These may take the form of adopting basic engineering design to prevent and/or minimise adverse landscape impact before adopting other mitigation or compensatory measures to alleviate the impacts. Potential mitigation measures should also include tree preservation and tree transplanting as far as practical, provision of buffer planting, reinstatement of affected landscape area, compensatory planting, erection of decorative screen hoarding that compatible with the surrounding setting, reprovision of amenity areas / open spaces, aesthetic design and layout of structures and façade treatment, and any measures to mitigate the impact on the existing and planned land uses and viewers. A programme for the mitigation measures is provided. The agencies responsible for the funding, implementation, management and maintenance of the mitigation measures are identified.

- ***Prediction of the significance thresholds of impacts before and after the implementation of the mitigation measures***. Potential mitigation measures suggested would alleviate the landscape impact and enhance the landscape quality by reinstating the disturbed lands and improve the compatibility with the surrounding. By synthesising the magnitude of the change and the sensitivity of the various LRs and LCAs, it is possible to categorise impacts in a logical, well-reasoned and consistent fashion. **Table 10.1** shows the rationale for dividing the degree of significance into four thresholds, namely negligible, slight, moderate, and substantial, depending on the combination of a negligible-small-intermediate-large magnitude of change and a low-medium-high degree of sensitivity of landscape resource and character.

Table 10.1 Relationship between Sensitivity and Magnitude of Change in Defining Impact Significance

Magnitude of Change	Low	Medium	High
Substantial	Moderate	Moderate / Substantial	Substantial
Moderate	Slight / Moderate	Moderate	Moderate / Substantial
Slight	Negligible / Slight	Slight / Moderate	Moderate
Negligible	Negligible	Negligible	Negligible

Sensitivity of Landscape Resource and Landscape Character Area

Note: All impacts are Adverse unless otherwise noted with Beneficial.

- The significance thresholds of landscape impact are categorised as follows:-

Substantial: Adverse impact where the proposal would cause significant deterioration or improvement in existing landscape quality.

Moderate: Adverse impact where the proposal would cause a noticeable deterioration or improvement in existing landscape quality.

Slight: Adverse impact where the proposal would cause a barely perceptible deterioration or improvement in existing landscape quality.

Negligible: No discernible change in the existing landscape quality.

Beneficial: Beneficial impact where the proposal would complement the landscape character of its setting, and/or would improve overall landscape quality.

Visual Impact Assessment Methodology

10.4.2 Reference is drawn from EIAO GN No. 8/2023 – Preparation of Landscape and Visual Impact Assessment Under Environmental Impact Assessment Ordinance.

10.4.3 These various elements of the visual impact assessment are detailed below:-

- ***Identification of the Visual Envelope of the Project.*** This is achieved by site visit and desktop study of topographic maps and photographs. Size, distance and other factors will be considered, to determine the Visual Envelope of the Project. The Visual Envelope is expected to cover the fields of views from sensitive viewers in direct sight of the proposed works; usually defined by natural ridgeline, man-made features, or road infrastructures, etc.

- ***Identification of the Key Public Viewing Points within the Visual Envelope.*** These viewing points are where members of the public or tourists can assess or view the site easily.

- ***Assessment of the degree of sensitivity to change of the viewers of the Viewing Points.*** Factors considered include:-

- Visual impact on sensitive public viewers from the most affected viewing points, include key pedestrian nodes, popular areas used by the public or tourists for outdoor activities, recreation, rest, sitting-out, leisure, walking, sight-seeing, and prominent travel routes where travelers' visual attention may be caught by the Project;

- People engaged in working activities are regarded as less sensitive to the visual changes;

- Viewing point should be at human eye level for a realistic presentation of the views;

- Key public viewing points may refer to Chapter 11 on Urban design Guidelines in the Hong Kong Planning Standards and Guidelines (HKPSG), the Explanatory Statement of the relevant statutory plans, adopted outline development plans and layout plans, and completed planning studies available for public reference; and

- Local viewpoints should be determined with reference to the setting of the project and views of local significance

• The sensitivity of viewers of the viewing point is classified as follows:-

High: The viewers of the viewing point are highly sensitive to any change in their viewing experience.

Medium: The viewers of the viewing point are moderately sensitive to any change in their viewing experience.

Low: The viewers of the viewing point are only slightly sensitive to any change in their viewing experience.

• ***Identification of existing visual elements,*** this includes major physical structures, visual resources or attractors, and / or visual eyesores or detractors that currently exist or area known to be planned within the assessment area. Different visual elements may enhance, degrade or neutralise the overall visual impact of the development being assessed.

• ***Appraisal of visual changes.*** Visual changes may be positive or negative and they are not necessarily mutually exclusive:-

- **Visual Composition**: the total visual effects of all the visual elements due to their variation in locations, massing, heights, dispositions, scales, forms, proportions and characters viz-a-viz the overall visual backdrop. It may result in visual balance, compatibility, harmony, unity or contrast. This appraisal should have due regard to the overall visual context and character within the wider and local contexts;

- **Visual Obstruction**: this appraisal should assess the degree of visual obstruction and loss of views or visual openness due to the Project from all key public viewing points within the assessment area. Blockage or partial blockage of views which substantially reduce visual permeability, existing panorama, vistas, visual resources or visual amenities should be avoided or minimised, in particular with regard to impact on prominent ridgelines, the harbour, natural coastlines, open sea horizon, skyline,

scenic areas, valued landscape, special landmark, heritage features to be preserved, etc; and

- **Visual Changes**: this appraisal should assess the impacts on changes with direct sightlines (considering degree of visibility and viewing distance) to the existing and future public views by comparing before and after the proposed works.

- The magnitude of visual changes is classified as follows:-

  Substantial: The viewers of the viewing point would suffer a major change in their viewing experience.

  Moderate: The viewers of the viewing point would suffer a moderate change in their viewing experience.

  Slight: The viewers of the viewing point would suffer a small change in their viewing experience.

  Negligible: The viewers of the viewing point would suffer no discernible change in their viewing experience.

- ***Identification of potential visual mitigation measures***. These may take the form of adopting basic engineering design to prevent and/or minimise adverse visual impact before adopting other mitigation or compensatory measures to alleviate the impacts. Potential mitigation measures should also include tree preservation and tree transplanting as far as practical, provision of buffer planting, reinstatement of affected landscape area, compensatory planting, erection of decorative screen hoarding that compatible with the surrounding setting, reprovision of amenity areas / open spaces, aesthetic design and layout of structures and façade treatment, and any measures to mitigate the impact on the existing and planned land uses and viewers. A programme for the mitigation measures is provided. The agencies responsible for the funding, implementation, management and maintenance of the mitigation measures are identified.

- ***Prediction of the significance thresholds of visual impact after the implementation of the mitigation measures***. By synthesising the magnitude of the various visual impact and the sensitivity of the viewers, it is possible to categorise the degree of significance of the impacts in a logical, well-reasoned and consistent fashion. **Table 10.2** shows the rationale for dividing the degree of significance into four thresholds, namely, negligible, slight, moderate and substantial, depending on the combination of a negligible-slight-moderate-substantial magnitude of change and a low-medium-high degree of sensitivity of viewers. The assessment of visual impacts will be presented in a matrix format considering the factors including the location of the viewing points, type and approximate number of viewers of the viewing points, description of existing view and degree of visibility of the proposed works, the sensitivity of the viewers of the viewing points, sources of visual impacts, minimum viewing distance of the viewers, magnitude of change, mitigation measures and significance thresholds of visual impacts (after mitigation measures established) at the operational phase.

Table 10.2 Relationship between Viewing Points / Viewers' Sensitivity and Magnitude of Change in Defining Impact Significance

Magnitude of Change	Low	Medium	High
Substantial	Moderate	Moderate / Substantial	Substantial
Moderate	Slight / Moderate	Moderate	Moderate / Substantial
Slight	Negligible / Slight	Slight / Moderate	Moderate
Negligible	Negligible	Negligible	Negligible

Sensitivity of Viewing Points / Viewers

Note: All impacts are Adverse unless otherwise noted with Beneficial.

- The significance thresholds of visual impact are categorised as follows:-

  Substantial: Adverse impact where the proposal would cause significant deterioration or improvement in existing visual quality.

  Moderate: Adverse impact where the proposal would cause a noticeable deterioration or improvement in existing visual quality.

  Slight: Adverse impact where the proposal would cause a barely perceptible deterioration or improvement in existing visual quality.

  Negligible: No discernible change in the existing visual quality.

  Beneficial: Beneficial impact where the proposal would complement the visual character of its setting, and/or would improve overall visual quality.

10.5 Baseline Study

Physical Landscape Resources

Landform

10.5.1 The assessment area for on-site works areas generally comprises various types of landforms, from developed area at northeast tip of the Loop, marsh and lowland grassland at the southern side of Hoo Hok Wai and northern side of Ma Tso Lung, to the hillside upland at the northern side of Tse Koo Hang and western side of Tai Shek Mo.

Water Body

10.5.2 Within the assessment area, streams and water body is covered, including natural river course bounding the Loop and in Ma Tso Lung, and fishponds at eastern side of Lok Ma Chau (LMC) and southern side of Hoo Hok Wai.

Existing Trees

10.5.3 A broad brush tree and vegetation survey was carried out to identify the existing trees and vegetation located within the project area and account the baseline condition of existing landscape resources at the time of EIA preparation. To quantify and safeguard the quality of broad brush tree and vegetation survey information, on-site survey at the nearest accessible areas in combination of aerial photo analysis and drone technology was adopted. For accessible areas (such as roadside areas of LMC and Ma Tso Lung), the tree and vegetation survey information was collected by on-site visual inspection. For remaining areas with site constraints and safety/accessibility problems (such as hillside locations and steep slopes, within private lots and fence off areas etc.), the tree quantity and tree information estimation were collected from aerial photos and drone photos of high resolution, with visual inspection at the nearest locations to quantify the findings as far as possible. Furthermore, existing trees would be carefully grouped in consideration of the extent of each LR, within site/ outside site location, tree and vegetation composition and density, accessibility and geographical location of the tree group etc. Therefore, the conducted on-site broad brush tree and vegetation survey, in combination of aerial photo and drone technology to increase the accuracy, provides representative tree and vegetation information for effective analysis.

10.5.4 Based on the broad brush tree and vegetation survey information, approximately 2,642 nos. of trees were surveyed within the 100m assessment area. None of these are Registered Old and Valuable Trees (OVTs). Dominant tree species include *Macaranga tanarius* var. *tomentosa, Celtis sinensis, Dimocarpus longan, Ficus microcarpa, Pinus elliottii* and *Melia azedarach* etc. that commonly found in Hong Kong were identified. 5 nos. of Trees of Particular Interest (TPIs) are identified in accordance with Section 3 of Appendix I of the EIA Study Brief No. ESB-375/2025. For the 5 nos. of TPIs, 1 no. of *Cinnamomum camphora* is in large size with DBH over 1m, and 4 nos. of *Aquilaria sinensis* with conservation interest under Cap. 586 are surveyed. A few numbers of undersized *Aquilaria sinensis* are also recorded in Chapter 8 of EIA Report, the indicative location also reflected in the broad brush tree and vegetation survey. The broad brush tree and vegetation survey including tree and vegetation survey plans and tree schedule are illustrated in **Appendix 10.1**.

Landscape Resources

10.5.5 Baseline landscape resources (LR) which will be potentially affected by the Project are described and appraised, together with their sensitivity are analysed and evaluated in **Table 10.3**. The locations of LR are mapped in Figures **60722948/A10/FIGURE 10.15** to **10.21**. Photographs of the LR are provided in Figures **60722948/A10/FIGURE 10.22** to **10.23**.

Table 10.3 Landscape Resources and Their Sensitivity

ID No.	Landscape Resources	Description	Sensitivity	Approx. Area (ha)
LR1	Natural Stream / River Course	This LR refers to natural streams and river course and its riparian vegetation, including the natural river course bounding the Loop to the east, the natural stream in Ping Hang and the natural stream course at Ma Tso Lung extends from the uplands of LMC ridge crossing the grassland of the valley floor before joining the fishpond landscapes in the northeast of the assessment area. The	High	2.98

ID No.	Landscape Resources	Description	Sensitivity	Approx. Area (ha)
		majority associated vegetation includes a combination of wetland herbs, common and weedy grass vegetation and some isolated shrub. Considering the natural characteristics of this LR, where the natural river course bounding the Loop forms a regionally significant landscape resource and a distinctive feature within the assessment area, and the natural streams integrates with the regionally significant fishponds and grassland landscape, the quality and significance of this LR is relatively high. This LR less susceptible to change and the ability to accommodate change is low, and hence the sensitivity is considered as High.		
LR2	Engineered Water Channel	This LR refers to the water channels have been engineered or channelised as part of drainage improvements and its riparian vegetation, including the constructed water channel at Ma Tso Lung located to the north and east of Tse Koo Hang, to provide flood relief for the agricultural valley floor. The common riparian plants found on the concrete banks of the engineered water channels. Considering the recent nature of the construction works, the quality and significance of this LR is relatively low and have a high ability to accommodate change, and hence the sensitivity is considered as Low.	Low	0.34
LR3	Fishponds	This LR mainly includes fishponds on the east of LMC and south of Hoo Hok Wai. These fishponds are characterised by their rectilinear shapes and are considered to be a regionally significant landscape resource and a distinctive feature within the assessment area. In general, the fishponds are interspersed with marshland areas where the water bodies have succumbed to natural succession. The earth bunds which separate the ponds are vegetated with a combination of coarse grassland and small groups and trees and shrubs. The main species composition of these areas includes *Macaranga tanarius* var. *tomentosa, Melia azedarach and Carica papaya* with some largely self-seeded trees such as *Leucaena leucocephala* and *Melia azedarach*. This LR	High	16.04

ID No.	Landscape Resources	Description	Sensitivity	Approx. Area (ha)
		is considered to be important to the landscape setting of the assessment area the quality and significance of these fishponds are high with relatively little tolerant to change. Hence, the sensitivity is considered as High.		
LR4	Marsh / Wetland	This LR refers to the marshland / wetland areas / seasonally wet grassland located in Hoo Hok Wai and Ma Tso Lung. These areas form a successional stage in the return of disused fishponds to dry land and represent a loss in the distinctive landscape texture of the organically shaped fishponds. Some self-seeded trees spreading from adjacent woodland located at the southern edge of the marsh near the Border Road. The main species composition of these areas includes *Dimocarpus longan, Ficus hispida* and *Clausena lansium* with some self-seeded trees spreading from adjacent woodland located at the southern edge of the marsh near the Border Road including species *Macaranga tanarius* var. *tomentosa*. This marsh area is considered to be a locally important LR and owing to factors such as its size, the nature of the vegetation and its appearance, the quality and significance of these marshlands are high with relatively low tolerant to change. Hence, the sensitivity of this LR is considered as High.	High	7.68
LR5	Man-made Marsh / Wetland	This LR mainly refers to the man-made marshland / wetland areas at the southern edge of the Loop as compensatory wetlands for part of the ecological mitigation measures to the Loop development. The quality and significance of this landscape resources are fair and have a medium ability to accommodate change, and hence the sensitivity is considered as Medium.	Medium	1.58
LR6	Mixed Woodland	This LR is the vegetated hillside, including mainly woodland with some exotic plantation species identified, located to the west of Shun Yee San Tsuen at Ma Tso Lung. It was fragmented by developed areas such as the Border Road, Ma Tso Lung and nearby villages. It is composed of woodland trees and areas of scrubland with the main species being *Celtis sinensis, Bombax ceiba* and *Melia azedarach*. 5 nos. of TPIs, including 1 no. of *Cinnamomum camphora* in	High	5.33

ID No.	Landscape Resources	Description	Sensitivity	Approx. Area (ha)
		large size with DBH over 1m, and 4 nos. of *Aquilaria sinensis* with conservation interest, are identified in this LR. The condition of the existing trees within this area are fair with more diverse native tree species and contribute to the local rural landscape Given the contribution of this LR to the landscape setting and backdrop to Ma Tso Lung, with its high amenity value and its importance to the local context and local importance, the quality and significance of this landscape resources are high with relatively low tolerance to change. The ability to accommodate change is relatively low. Hence, the sensitivity of this LR is considered as High.		
LR7	Grassland / Shrubland	This LR mainly includes the grassland and shrubland located on the hill slopes in Ma Tso Lung, to the west of Tai Shek Mo, and area near the HKPF LMC Operation Base, with small portion of plantation at the foot of Tai Shek Mo. The shrubland forms the transitional landscape between the wooded lower hill slopes and extends to the grassland at the summit of the hills, forming part of the green backdrop to many of the views available in locations. The main species composition of these areas includes *Pinus elliottii, Celtis sinensis* and *Ficus microcarpa,* which are spreading from adjacent woodlands or cultivated by villagers in settlement along sides of existing Border Road and at the lower slopes of Tai Shek Mo. This LR is in combination with the other hillside LRs and forming part of the backdrop to views of the landscape from lower levels. The quality and significance of this LR is medium and have medium tolerance to change, hence the sensitivity is considered as Medium.	Medium	21.01
LR8	Agricultural Fields	This LR refers to the agricultural fields to the east of the village of Ma Tso Lung, which are located on the valley floor between the lower hill slopes of Ma Tso Lung and Tai Shek Mo Some largely fruit trees such as *Carica papaya* and *Musa acuminata 'Dwarf Cavendish' indica* are found within or at the edge the agricultural fields. This area is formed from both active and abandoned agricultural land, and is relatively important	Medium	0.13

ID No.	Landscape Resources	Description	Sensitivity	Approx. Area (ha)
		to the landscape setting of the adjacent village houses. The quality and significance of this LR is fair, and its ability to accommodate change is medium. Hence, the sensitivity is considered as Medium.		
LR9	Vegetation in Village Settlement	This LR refers to vegetation in village settlement, including Tse Koo Hang Village Settlement and Ma Tso Lung Village Settlement. For this LR in Tse Koo Hang Village Settlement, it is composed temporary squatters adjacent to abandoned agriculture fields which are largely abandoned and was originally utilised for cultivation and framing purpose; in Ma Tso Lung Village Settlement, it is composed of an intermediate number of three-storey village houses and temporary squatters adjacent to agriculture fields, which the houses and structures are located on a series of terraced platforms extending to the mid-slopes of Crest Hill. The main species found within this LR includes *Dimocarpus longan, Ficus microcarpa* and *Macaranga tanarius* var. *tomentosa*. The condition and amenity value of the existing trees within this area are fair, the trees are generally young to mature and contribute to the amenity value and the local rural landscape. Considering the quality and significance of this LR is fair and its medium tolerance to accommodate change, the sensitivity is considered as Medium.	Medium	12.71
LR10	Vegetation in Developed Area	This LR refers to the vegetation found in developed area and along roads corridors, including the Central Government-Aided Emergency Hospital located in the Loop and the HKPF LMC Operation Base adjacent to Border Road. The road corridor is lined by tree planting and secondary woodland however it has little intrinsic value as an LR. The vegetation found mainly the ruderal herbs and weedy self-seeded trees such as *Leucaena leucocephala* or some individual trees scattered at the edges of the development plots. The quality of these vegetation is fair to poor, while significance is relatively low and the ability to accommodate change is high. Hence, the sensitivity is considered as Low.	Low	12.30

ID No.	Landscape Resources	Description	Sensitivity	Approx. Area (ha)
LR11	Roadside Planting	This LR refers to the vegetation planted on the engineered slope near the Border Road near Hoo Hok Wai and Ma Tso Lung. The majority of the vegetation are slope planting, complies with cluster of exotic trees species and shrub and ruderal herbs and weedy. It is extensively covered throughout the roadside engineering slopes to provide a series of green screening and a green buffer between the roads and the adjacent hillside vegetation. Dominant tree species include *Lagerstroemia speciosa*, *Litsea glutinosa* and *Schefflera heptaphylla* . Its quality and significance of this man-made landscaping are relatively low while the ability to accommodate change is high. Hence, the sensitivity is considered as Low.	Low	0.64

Landscape Character Area

10.5.6 LCAs were identified within the assessment area in accordance with the "A Study on Landscape Value Mapping of Hong Kong". LCAs which will be potentially affected by the Project are described and appraised, together with their sensitivity analysed and evaluated in **Table 10.4**. The locations of LCAs are mapped in **60722948/A10/FIGURE 10.24** to **10.30**. Photographs of the LCAs are presented in **60722948/A10/FIGURE 10.31**.

Table 10.4 Landscape Character Areas and Their Sensitivity

ID No.	Landscape Character Area	Descriptions	Sensitivity	Approx. Area (ha)
LCA 1	Miscellaneous Urban Fringe Landscape of the Loop	This LCA refers to the development area in the Loop, including the works area to be developed into the HSITP, a portion of area temporarily constructed as the Central Government Aided Emergency Hospital and the ecological area created to south of the Loop. Vegetation found mainly man-made either for amenity purpose or greenery provision. Its quality and significance are considered as low while the ability to accommodate change is high. Hence, the sensitivity is considered as Low.	Low	11.84
LCA 2	Lowland Rural Landscape of Lok Ma Chau	This LCA is characterised predominantly by fishponds, scattered village house at Ping Hang and vegetated lower slopes in the background to the south of the Border Road. The rural landscape of this area forms the	Medium	9.59

ID No.	Landscape Character Area	Descriptions	Sensitivity	Approx. Area (ha)
		counterpoint to the urban setting of Shenzhen to the north. Vegetation is dominated by shrubby grassland and patches of plantation. Matures vegetation may found along the foothills. Its quality and significance are considered as fair while the ability to accommodate change is medium. Hence, the sensitivity is considered as Medium.		
LCA 3	Hillside Landscape of Lok Ma Chau	This LCA refers to hillside landscape of low ridges extending from LMC to Tse Koo Hang. Vegetation in the higher elevations of these ridges are characterised by a tapestry of shrub and coarse grassland, while in the lower elevations adjacent to villages and fishponds are dominated by mixed woodland. It forms the green backdrop to the viewers in the lowland areas of the Loop and Hoo Hok Wai. This LCA also contrasts with the dense urban development to the north of the Shenzhen River and form an important landscape feature. This LCA is important and significant to the adjacent district. Its quality is considered as high while the ability to accommodate change is low. Hence, its sensitivity is considered as High.	High	11.37
LCA 4	Lowland Rural Landscape of Hoo Hok Wai	This LCA is characterised predominantly by fishponds and marshlands which are major landscape components of the natural lowland character extending from Deep Bay to Mai Po to Sam Po Shue and Hoo Hok Wai. The marshes in this area have naturally evolved from abandoned fishponds or agricultural fields. Following the decline of agricultural activity in the territory since 1980, the area of marshland is expanding into Ma Tso Lung. The northern edge of Hoo Hok Wai was largely disturbed by the river training works for the Shenzhen River and a maintenance access for the river channel and forms the only developed element within this natural setting. The quality of this LCA is relatively high and the ability to accommodate change is relatively low. Given the local significance of these marshes and their unique lowland character extending from the landscape of	High	17.45

ID No.	Landscape Character Area	Descriptions	Sensitivity	Approx. Area (ha)
		LMC, it is considered to have a high sensitivity.		
LCA 5	Lowland Rural Landscape of Ma Tso Lung	This LCA refers to the scattered village house clusters and their network of associated agricultural fields and marsh / wetland in Ma Tso Lung, including Ma Tso Lung Shun Yee San Tsuen, Ma Tso Lung San Tsuen and village settlements in Tse Koo Hang. Vegetation is dominated by shrubby grassland and patches of plantation, with some marshes in Ma Tso Lung to the west of village settlements which have naturally evolved from former fishponds and agricultural fields. Matures vegetation may found along the foothills. Its quality is considered as fair while the ability to accommodate change is medium. Given the significance of its rural landscape character forming a buffer to Hoo Hok Wai wetland area, the sensitivity is considered as Medium.	Medium	23.09
LCA 6	Hillside Landscape of Ma Tso Lung	This LCA refers to hillside landscape of low ridges extending from Hoo Hok Wai to Tai Shek Mo. Vegetation in the higher elevations of these ridges are characterised by a tapestry of shrub and coarse grassland. It forms the green backdrop to the viewers in the lowland areas of Ma Tso Lung. This LCA also contrasts with the dense urban development to the north of the Shenzhen River and form an important landscape feature. This LCA is important and significant to the adjacent district. Its quality is considered as high while the ability to accommodate change is low. Hence, its sensitivity is considered as High.	High	7.41

Visual Envelope

10.5.7 The visual envelope of the Project has been identified by desktop study maps, photographs and site visit to determine visibility of the project site from various locations. Subject to the location of the proposed works which is in between the Loop, Hoo Hok Wai and Ma Tso Lung, the visual envelope is relatively extensive, confined along riverfront along the Loop and Hoo Hok Wai at the north, the ridgeline of the hillside along Ma Tso Lung at the east, the ridgeline of the hillside along the LMC at the south and the road infrastructure of LMC Control Point at the west. The visual envelope of the Project during operational phase is illustrated in

60722948/A10/FIGURE 10.32 and will be further reviewed subject to the latest design development.

Existing Visual Condition and Key Visual Elements

10.5.8 Within the visual envelope, most of them are lowland areas and ridgeline of the surrounding hillside of Ma Tso Lung and LMC. The major visual resource and landmark within the visual envelope is the marsh/wetland in Ma Tso Lung and fishponds in LMC and Hoo Hok Wai, continuous ridgeline with dense vegetation as greenery backdrop of most of the key public view and vast open sky view.

Existing/planned/approved Landuses as Baseline Conditions within Visual Envelope

10.5.9 There are 3 concurrent projects are identified within the visual envelope as shown in **60722948/A10/FIGURE 10.32**. The approved scheme of "Development of the Loop" in the Approved EIA Report [Register No.: AEIAR-176/2013] has been incorporated as baseline condition for the visual impact assessment. While for the other 2 concurrent projects within the visual envelope which are still under planning, namely "Ma Tso Lung Area and Other Sites in Kwu Tung North New Development Area and North District – Zone 1 Ma Tso Lung Area" and "Northern Metropolis Highway – Kwu Tung Section", there are not sufficient available design information obtained during the course of the EIA Study for incorporation in the baseline condition for the visual impact assessment.

Identified Public Viewing Points (VPs)

10.5.10 Within the visual envelope, the following key public VPs are identified.

1) VP1 – View from Ma Tso Lung Fishpond;

2) VP2 – View from Ma Tso Lung Eco Park;

3) VP3 – View from Macintosh Fort (Ma Tso Lung);

4) VP4 – View from footpath of Ma Tso Lung San Tsuen;

5) VP5 – View from She Leng towards Ma Tso Lung; and

6) VP6 – View from She Leng towards Hoo Hok Wai

10.5.11 Based on the location of the proposed aboveground structure, the viewpoints from key public viewers are mapped in **60722948/A10/FIGURE 10.32**, with their description and sensitivities evaluated in **Table 10.5** as follow:-

Table 10.5 Key Public Viewing Points and Their Sensitivity

VP. ID	Location of VPs and Description	Type of Viewers	Description and Value of Existing Views	Viewer Sensitivity (Low, Medium, High)
VP1	View from Ma Tso Lung fishpond • Ground level VP orientated towards ECR • Located at around 250m from the Project • Approx. no. of viewers: Few	Recreational / Traveller	- The general view of this VP consists of an open sky with the LMC mountain background. The Shenzhen skyline and the future development of the Loop are visible in the distant background. The foreground contains fishpond, adjacent greenery and shrubs. - The duration of view is short to medium, depending on whether viewers will pause at the point of the trail to view the mountain and Shenzhen skyline. - Value of existing view: Good	Medium
VP2	View from Ma Tso Lung Eco Park • Ground level VP orientated towards ECR • Located at around 80m from the Project • Approx. no. of viewers: Medium	Recreational / Traveller	- The general view of this VP consists of an open sky and the Ma Tso Lung mountains background. The foreground is a fishpond with adjacent greenery bound by the existing LMC Border Road. - The duration of view is short, as most viewers along the surrounding roads are transient. - Value of existing view: Fair	Medium
VP3	View from Macintosh Fort (Ma Tso Lung) • Elevated vantage point orientated towards ECR • Located at around 170m from the Project • Approx. no. of viewers: Few	Recreational (Hikers)	- The general view of this VP consists of an open sky and mountainous backdrop. There are a few village houses in Ma Tso Lung and She Leng with marsh/wetland are visible in the mid-ground. Meanwhile, the foreground is characterised by the dense vegetation on the green mountain slope. - The duration of view is short to medium, as most viewers along the road to Macintosh Fort are transient	Medium

VP. ID	Location of VPs and Description	Type of Viewers	Description and Value of Existing Views	Viewer Sensitivity (Low, Medium, High)
			and may stop for short period of time to enjoy the views of vast landscape. - Value of existing view: Good	
VP4	View from footpath of Ma Tso Lung San Tsuen • Ground level VP orientated towards ECR • Located at around 300m from the Project • Approx. no. of viewers: Few	Recreational / Traveller	- The general view of this VP consists of an open sky and the mountain backdrop of Lei Yue Shan behind Ma Tso Lung Shun Yee San Tsuen. There are a few village houses at the foot of the mountain visible through the trees and thick shrubs. There is a sprawling marsh/wetland with overgrown vegetation in the foreground. - The duration of view is short, as most viewers along the path are transient. - Value of existing view: Fair	Low
VP5	View from She Leng towards Ma Tso Lung • Elevated vantage point orientated towards the ECR • Located at around 750m from the Project • Approx. no. of viewers: Few	Recreational (Hikers)	- The general view of this VP consists of an open sky and the mountain backdrop behind Ma Tso Lung Shun Yee San Tsuen. There are a few village houses at the foot of the mountain visible through the trees and thick shrubs in the mid-ground. Meanwhile, the foreground is characterised by the dense shrubs and groundcovers on the green mountain slope. - The duration of view is short to medium, as most viewers along the walking trail in She Leng are transient and may stop for short period of time to enjoy the views of vast landscape. - Value of existing view: Good	High

VP. ID	Location of VPs and Description	Type of Viewers	Description and Value of Existing Views	Viewer Sensitivity (Low, Medium, High)
VP6	View from She Leng towards Hoo Hok Wai • Elevated vantage point orientated towards the ECR • Located at around 130m from the Project • Approx. no. of viewers: Few	Recreational (Hikers)	- The general view of this VP consists of the Shenzhen skyline at the background, the future development of the Loop and Hoo Hok Wai fishponds under an open sky at the mid-ground and framed by the green mountain slopes of She Leng at the foreground. - The duration of view is short to medium, as most viewers along the walking trail in She Leng are transient and may stop for short period of time to enjoy the views of the Shenzhen skyline and vast landscape. - Value of existing view: Good	High

10.6 Landscape Impact Assessment

Sources of Landscape Impact

10.6.1 The sources of landscape impact arising from the Project would create various levels of landscape impact during construction and operational phases of the Project. Potential impacts would result from the temporary and permanent works during the construction phase and permanent aboveground structure elements during operational phase.

10.6.2 The ECR will be predominantly as comprises the construction of connection roads (in form of depressed road, underpass and viaduct) connecting the eastern part of the Loop to KTN NDA via Ma Tso Lung Area. The sources of temporary landscape impact were expected from the following key aboveground construction works:-

- Construction of single 4-lane two-way carriageway, in form of at-grade road, depressed road, underpass and viaduct, of a total length of approximately 2.5km;
- Provision of E&M plant room and drainage pumping room to support the operation of depressed road and underpass;
- Modification of existing Border Road; and
- Associated street furniture, traffic aids, drainage, drainage, sewerage, water, utilities, lightings, electrical and mechanical works, slopes, retaining walls and landscaping works.

10.6.3 The sources of landscape impact in the operational phase would include:-

- Operation of the ECR, including carriageway and associated structures, etc.; and
- Permanent removal of existing trees and other vegetation.

Magnitude of Landscape Change

10.6.4 The magnitude of unmitigated landscape changes associated with the construction and operational phases of the Project were assessed and are described in **Table 10.6**.

Table 10.6 Magnitude of Landscape Changes during Construction and Operation

ID No.	Landscape Resources/ Landscape Character Areas	Description of Works	Description of Impacts	Magnitude of Change* (Negligible / Slight / Moderate / Substantial)	
				Construction	Operation
Landscape Resources					
LR1	Natural Stream / River Course	• No temporary work or permanent work proposed.	• Nil.	Negligible	Negligible

ID No.	Landscape Resources/ Landscape Character Areas	Description of Works	Description of Impacts	Magnitude of Change* (Negligible / Slight / Moderate / Substantial)	
				Construction	Operation
LR2	Engineered Water Channel	• No temporary work or permanent work proposed.	• Nil.	Negligible	Negligible
LR3	Fishponds	• Temporary work sites for the construction works of the Eastern Connection Road, including construction of underpass and associated structures. • Permanent structure of associated structure at the east of LMC and south of Hoo Hok Wai.	• Approx. 0.13 ha (0.89%) would be affected • A small portion of the LR that close to the works site would be temporarily affected. • A relatively small amount of existing shrub and weedy plant would be permanently removed.	Slight	Slight
LR4	Marsh / Wetland	• Temporary work sites for the construction works of the Eastern Connection Road, including construction of underpass, at-grade carriageway and associated structures. • Permanent structure of at-grade carriageway and ancillary works e.g. slope works and landscaping works at the east of LMC and south of Hoo Hok Wai.	• Approx. 1.02 ha (13.22%) would be affected. • A small portion of the LR that close to the works site would be temporarily affected. • Relatively small amount of existing trees would be affected.	Slight	Slight
LR5	Man-made Marsh / Wetland	• No temporary work or permanent work proposed	• Nil.	Negligible	Negligible
LR6	Mixed Woodland	• Temporary work sites for the construction works of the Eastern Connection Road, including slope cutting, construction of at-grade carriageway and associated structures.	• Approx. 0.97 ha (18.2%) would be affected. • Some of the existing vegetation would be affected permanently, especially the	Moderate	Moderate

ID No.	Landscape Resources/ Landscape Character Areas	Description of Works	Description of Impacts	Magnitude of Change* (Negligible / Slight / Moderate / Substantial)	
				Construction	Operation
		Permanent structure of at-grade carriageway and ancillary works e.g. slope works and landscaping works at the east of LMC and south of Hoo Hok Wai.	vegetation north of Ma Tso Lung. • 4 nos. of TPIs *Aquilaria sinensis* would be affected. • Existing natural hillside would be replaced by man-made slope.		
LR7	Grassland / Shrubland	• Temporary work sites for the construction works of the Eastern Connection Road, including slope cutting, construction of underpass, depressed road, at-grade and viaduct carriageway and associated structures at east of LMC and north of Ma Tso Lung. • Permanent structure of depressed road, at-grade and viaduct carriageway and ancillary works e.g. slope works and landscaping works at the east of LMC and north of Ma Tso Lung.	• Approx. 1.75 ha (8.33%) would be affected. • Some of the existing vegetation would be affected permanently, especially the vegetation at north of Ma Tso Lung • Existing grassland / shrubland would be replaced by man-made slope.	Moderate	Moderate
LR8	Agricultural Fields	• No temporary work or permanent work proposed.	• Nil.	Negligible	Negligible
LR9	Vegetation in Village Settlement	• Temporary work sites for the construction works of the Eastern Connection Road, including construction of at-grade and viaduct carriageway and associated structures at the north of Ma Tso Lung. • Permanent structure of at-grade and viaduct carriageway at the north of Ma Tso Lung.	• Approx. 0.18 ha (1.42%) would be affected. • Some of the existing vegetation in village settlements would be removed permanently.	Slight	Slight

ID No.	Landscape Resources/ Landscape Character Areas	Description of Works	Description of Impacts	Magnitude of Change* (Negligible / Slight / Moderate / Substantial)	
				Construction	Operation
LR10	Vegetation in Developed Area	• Temporary work sites for the construction works of the Eastern Connection Road, including construction of depressed road at the Loop, at-grade and viaduct carriageway and associated structures at the north of Ma Tso Lung. • Permanent structure of depressed road, at-grade carriageway and ancillary works e.g. landscaping works, at the Loop and north of Ma Tso Lung.	• Approx. 0.64 ha (5.23%) would be affected. • Some of the existing vegetation in developed area would be removed permanently.	Slight	Slight
LR11	Roadside Planting	• Temporary work sites for the construction works of the Eastern Connection Road, including slope cutting, construction of underpass, at-grade carriageway and associated structures at east of LMC. • Permanent structure of at-grade carriageway and ancillary works e.g. slope works and landscaping works at the east of LMC.	• Approx. 0.06 ha (9.58%) would be affected. • Some of the existing vegetation along the engineering slope would be removed.	Slight	Slight
Landscape Character Area					
LCA 1	Miscellaneous Urban Fringe Landscape of the Loop	• Temporary work sites for the construction works of the Eastern Connection Road, including construction of depressed road, underpass and associated structures. • Permanent structure of depressed road and associated structure.	• Approx. 0.46 ha (3.87%) would be affected. • A small portion of the LCA would be permanently taken up by proposed works and replaced by the transportation related structure (i.e. depressed road and associated structure).	Slight	Slight

ID No.	Landscape Resources/ Landscape Character Areas	Description of Works	Description of Impacts	Magnitude of Change* (Negligible / Slight / Moderate / Substantial)	
				Construction	Operation
LCA 2	Lowland Rural Landscape of Lok Ma Chau	• Temporary work sites for the construction works of the Eastern Connection Road, including construction of underpass and associated structures. • Permanent structure of associated structure.	• Approx. 0.13 ha (1.37%) would be affected. • A relatively small portion of the LCA would be permanently taken up by proposed works and replaced by the transportation related structure (i.e. associated structure).	Slight	Slight
LCA 3	Hillside Landscape of Lok Ma Chau	• Temporary work sites for the construction works of the Eastern Connection Road, including slope cutting, construction of depressed road, at-grade carriageway and associated structures. • Permanent structure of depressed road, at-grade carriageway, associated structure and ancillary works e.g. slope works.	• Approx. 0.71 ha (6.22%) would be affected. • A relatively small portion of hillside landscape would be permanently taken up by proposed works and replaced by the transportation related structure (i.e. depressed road, at-grade carriageway, and associated structure), and man-made slope.	Slight	Slight
LCA 4	Lowland Rural Landscape of Hoo Hok Wai	• Temporary work sites for the construction works of the Eastern Connection Road, including construction of at-grade carriageway and associated structures • Permanent structure of at-grade carriageway, associated structure and ancillary works e.g. slope works and landscaping works.	• Approx. 1.05 ha (6.01%) would be affected. • A relatively small portion of marsh/wetland would be permanently taken up by proposed works and replaced by the transportation related structure (i.e. at-grade carriageway and associated structure).	Slight	Slight

ID No.	Landscape Resources/ Landscape Character Areas	Description of Works	Description of Impacts	Magnitude of Change* (Negligible / Slight / Moderate / Substantial)	
				Construction	Operation
LCA 5	Lowland Rural Landscape of Ma Tso Lung	• Temporary work sites for the construction works of the Eastern Connection Road, including slope cutting, construction of at-grade and viaduct carriageway and associated structures • Permanent structure of at-grade and viaduct carriageway, associated structure and ancillary works e.g. slope works and landscaping works.	• Approx. 1.07 ha (4.63%) would be affected. • A small portion of the LCA would be permanently taken up by proposed works and replaced by the transportation related structure (i.e. at-grade and viaduct carriageway, and associated structure), and man-made slope.	Slight	Slight
LCA 6	Hillside Landscape of Ma Tso Lung	• Temporary work sites for the construction works of the Eastern Connection Road, including slope cutting, construction of at-grade and viaduct carriageway and associated structures • Permanent structure of at-grade and viaduct carriageway, associated structure and ancillary works e.g. slope works and landscaping works.	• Approx. 1.19 ha (16.05%) would be affected. • A moderate portion of the hillside landscape would be permanently taken up by proposed works and replaced by the transportation related structure (i.e. at-grade and viaduct carriageway, and associated structure), and man-made slope.	Moderate	Moderate

Significance of Unmitigated Landscape Impact

10.6.5 The significance of landscape impact, before implementation of mitigation measures, in the construction and operational phases will be assessed and presented in **Table 10.10**.

10.6.6 The initial significance of unmitigated impacts on the affected LRs and LCAs would vary from substantial to negligible. They are described as below.

LR 1 - Natural Stream / River Course

10.6.7 This LR refers to the natural streams and river course and its riparian vegetation, including the natural river course bounding the Loop to the east, the natural stream in Ping Hang and the natural stream course at Ma Tso Lung extends from the uplands

of LMC ridge crossing the grassland of the valley floor before joining the fishpond landscapes in the northeast of the assessment area. The sensitivity of this LR is identified as high. Since no works would be proposed during both construction and operational phases, hence the magnitude of impact on these LRs due to the Project is considered as **Negligible**. The resultant unmitigated impact during construction and operational phases is **Negligible**.

LR 2 – Engineered Water Channel

10.6.8 This LR refers to the water channels have been engineered or channelised as part of drainage improvements and its riparian vegetation, including the constructed water channel at Ma Tso Lung located to the north and east of Tse Koo Hang, to provide flood relief for the agricultural valley floor. The common riparian plants found on the concrete banks of the engineered water channels. The sensitivity of this LR is identified as low. Since no works would be proposed during both construction and operational phases, hence the magnitude of impact on these LRs due to the Project is considered as **Negligible**. The resultant unmitigated impact during construction and operational phases is **Negligible**.

LR 3 – Fishponds

10.6.9 This LR refers to fishponds on the east of LMC and south of Hoo Hok Wai. The sensitivity of this LR is considered as high due to its local importance. A periphery of this LR may potentially be affected the construction works of the Project at the east of LMC and south of Hoo Hok Wai during both the construction and operational phases. The changes would be irreversible while the compatibility of the proposed works is considered as low. The magnitude of change on this LR due to the Project is considered as **Slight** and localise. Given to its high sensitivity, the resultant unmitigated impact during construction and operational phases is **Moderate**.

LR 4 – Marsh / Wetland

10.6.10 This LR refers to the marshland / wetland areas located in Hoo Hok Wai and Ma Tso Lung within the Study Area. The sensitivity of this LR is considered as high due to its local importance and owing to factors such as its size, the nature of the vegetation and its appearance. Among the extensive marsh / wetland, only a small portion of the marsh/wetland at the east of LMC and south of Hoo Hok Wai would be affected by the proposed construction works, and relatively small amount of existing trees would be affected. The compatibility of the proposed works is fair while changes are irreversible. Considered that the proposed works are localised and relatively small in scale, the magnitude of impact on this LR due to the Project is considered as **Slight**. Given to its high sensitivity, the resultant unmitigated impact during construction and operational phases is **Moderate**.

LR 5 – Man-made Marsh / Wetland

10.6.11 This LR refer refers to the man-made marshland / wetland areas at the southern edge of the Loop as part of the ecological mitigation measures to the Loop development. The sensitivity of this LR is medium. Since no works would be proposed during both construction and operational phases, hence the magnitude of impact on these LRs due to the Project is considered as **Negligible**. The resultant unmitigated impact during construction and operational phases is **Negligible**.

LR 6 – Mixed Woodland

10.6.12 This LR refers to the vegetated slopes located to the west of Shun Yee San Tsuen at Ma Tso Lung. The sensitivity of this LR is high due to the fair condition, high amenity value and its importance to the local context and local importance. During the construction stage, only a small portion of vegetated area would be affected for the construction of the at-grade carriageways and associated structures. Although 4 nos. of TPIs (i.e. *Aquilaria sinensis*) will be unavoidably affected by the works, the proposed road alignment can preserve 1 no. of TPI in large size (i.e. *Cinnamomum camphora*) in situ. Upon completion of works, the disturbed natural hillside would be replaced as man-made slope permanently due to the associated slope cutting works and construction of highway structures. The compatibility of the proposed works is low to the surrounding natural landscape and change is irreversible. Considered that the proposed works are localised and relatively small in scale, the magnitude of impact on this LR due to the Project is considered as **Moderate**. Given to its high sensitivity, the resultant unmitigated impact during construction and operational phases is **Moderate**.

LR 7 – Grassland / Shrubland

10.6.13 This LR refers to grassland and shrubland located on the hill slopes in Ma Tso Lung, to the west of Tai Shek Mo, and area near the HKPF LMC Operation Base. The sensitivity of this LR is medium due to the fair quality and significance. During the construction stage, only a small portion of vegetated area would be affected for the construction of the depressed road, at-grade and viaduct carriageways, footpaths, and associated structures. Upon completion of works, the disturbed natural grassland / shrubland would be replaced as man-made slope permanently due to the associated slope cutting works and construction of highway structures. The compatibility of the proposed works is low to the surrounding natural landscape and change is irreversible. The magnitude of impact on this LR due to the Project is considered as **Moderate**. Given to its medium sensitivity, the resultant unmitigated impact during construction and operational phases is **Moderate**.

LR 8 – Agricultural Fields

10.6.14 This LR refers to the agricultural fields to the east of the village of Ma Tso Lung, which are located on the valley floor between the lower hill slopes of Ma Tso Lung and Tai Shek Mo. The sensitivity of this LR is identified as medium. Since no works would be proposed during both construction and operational phases, hence the magnitude of impact on these LRs due to the Project is considered as **Negligible**. The resultant unmitigated impact during construction and operational phases is **Negligible**.

LR 9 – Vegetation in Village Settlement

10.6.15 This LR refers to vegetation in village settlement, including Tse Koo Hang Village Settlement and Ma Tso Lung Village Settlement. The sensitivity of this LR is identified as medium. During the construction phase, it is assumed that some portion of existing vegetation would be affected for the construction of the at-grade and viaduct carriageway and associated structures. Upon the completion of works, some of the existing the vegetation in village settlements would be removed permanently. The compatibility of the proposed works is fair while the change is irreversible. The magnitude of impact on this LR due to the Project is considered as **Slight** and the resultant unmitigated impact during construction and operational phases is **Slight**.

LR 10 – Vegetation in Developed Area and Roads

10.6.16 This LR refers to the vegetation found in developed area and along roads corridors. The sensitivity of this LR is identified as low. During the construction phase, it is assumed that a small portion of existing the vegetation at the edges of the developed area would be removed of this LR for the construction of the depressed road, at-grade carriageway and associated structures. The affected area would be reinstated upon the completion of works. Considered that the sensitivity of this LR is low while the magnitude of impact on this LR due to the Project is considered as **Slight**, the resultant unmitigated impact during construction and operational phases is **Slight**.

LR 11 – Roadside Planting

10.6.17 This LR refers to the vegetation planted on the engineered slope near the Border Road near Hoo Hok Wai and Ma Tso Lung. The sensitivity of this LR is identified as low. Upon completion of works, it is assumed that a small portion of the existing vegetation would be removed permanently of this LR for the construction of at-grade carriageway and associated structures. The affected area would be reinstated upon the completion of works. Considered that the sensitivity of this LR is low while the magnitude of impact on this LR due to the Project is considered as **Slight**, the resultant unmitigated impact during construction and operational phases is **Slight**.

LCA 1 – Miscellaneous Urban Fringe Landscape of the Loop

10.6.18 This LCA refers to the developed area in the Loop, including reclaimed lowland with Central Government Aided Emergency Hospital and the ecological area created to south of the Loop. The sensitivity of this LCA is identified as low. During the construction stage, a small portion of the area would be permanently taken up by proposed works and replaced by the transportation related structure (i.e. depressed road, underpass, and associated structure). The affected area would be reinstated upon the completion of works. The compatibility of the proposed works is high while the changes are considered irreversible. Considered that the sensitivity of this LCA is low while the magnitude of impact on this LCA due to the Project is considered as **Slight**, the resultant unmitigated impact during construction and operational phases is **Slight**.

LCA 2 – Lowland Rural Landscape of Lok Ma Chau

10.6.19 During the construction stage, a small portion of the area such as marsh / wetland and fishponds would be affected for the construction works of the Project. Upon the completion of works, the vegetation would be permanently affected and occupied by the transportation related infrastructure and the affected vegetation would be reinstated or reprovided as form of roadside planting. The changes are irreversible, and compatibility of works is fair to low. Since the changes are localised and relatively small in scale, the magnitude of impact to this LCA is considered as **Slight**. Given to its medium sensitivity, the unmitigated landscape impact on this LCA is considered as **Slight** during construction and operational phases.

LCA 3 – Hillside Landscape of Lok Ma Chau

10.6.20 During the construction stage, the periphery of the existing lush and densely vegetated hillsides and the engineering slopes would be affected for the construction works of the Project. Upon the completion of works, the vegetation would be permanently affected and occupied by the transportation related infrastructure and

the affected vegetation would be reinstated or reprovided as form of roadside planting or man-made slope vegetation. The changes are irreversible, and compatibility of works is fair to low. Since the changes are localised and relatively small in scale, the magnitude of impact to this LCA is considered as **Slight**. Given to its high sensitivity, the unmitigated landscape impact on this LCA is considered as **Moderate** during construction and operational phases.

LCA 4 – Lowland Rural Landscape of Hoo Hok Wai

10.6.21 This LCA is characterised predominantly by fishponds and marshlands which are major landscape components of the natural lowland character extending from Deep Bay to Mai Po to Sam Po Shue and Hoo Hok Wai. Vegetation found mainly along the roadside or area between the existing road and Hoo Hok Wai fishponds with relatively fair amenity value. The sensitivity is considered as high due to its local importance. During the construction stage, periphery of this LCA may potentially be affected the construction works of the Project. Upon the completion of works, the vegetation would be permanently affected and occupied by the transportation related infrastructure and the affected vegetation would be reinstated or reprovided as form of roadside planting or man-made slope vegetation. The changes are irreversible, and compatibility of works is fair to low. Since the changes are localised and relatively small in scale, the magnitude of impact to this LCA is considered as **Slight**. Given to its high sensitivity, the unmitigated landscape impact on this LCA is considered as **Moderate** during construction and operational phases.

LCA 5 – Lowland Rural Landscape of Ma Tso Lung

10.6.22 This LCA refers to the scattered village house clusters and their network of associated agricultural fields and marsh / wetland in Ma Tso Lung, including Ma Tso Lung Chun Yee San Tsuen, Ma Tso Lung San Tsuen and village settlements in Tse Koo Hang. The sensitivity is considered as medium with fair quality and significance. During the construction stage, existing vegetation may potentially be affected the construction works of the Project. Upon the completion of works, the vegetation would be permanently affected and occupied by the transportation related infrastructure and the affected vegetation would be reinstated or reprovided as form of roadside planting or man-made slope vegetation. The changes are irreversible while compatibility of works is fair. The magnitude of impact to this LCA is considered as **Slight**. Given to its medium sensitivity, the unmitigated landscape impact on this LCA during construction and operational phase is considered as **Slight**.

LCA 6 – Hillside Landscape of Ma Tso Lung

10.6.23 During the construction stage, the existing lush and densely vegetated hillsides and the engineering slopes would be affected for the construction works of the Project. Upon the completion of works, the vegetation would be permanently affected and occupied by the transportation related infrastructure and the affected vegetation would be reinstated or reprovided as form of man-made slope vegetation. The changes are irreversible, and compatibility of works is fair to low. The magnitude of impact to this LCA is considered as **Moderate**. Given to its high sensitivity, the unmitigated landscape impact on this LCA during construction and operational phase is considered as **Moderate**.

10.7 Visual Impact Assessment

Sources of Visual Impact

10.7.1 The sources of visual impact in operational phase of the Project would create varying levels of visual impact. Potential impacts would be resulted from the permanent aboveground structure elements during operational phase.

10.7.2 The sources of visual impact in the operational phase would include:-

- Operation of single 4-lane two-way carriageway, in form of at-grade road, depressed road, underpass and viaduct, of a total length of approximately 2.5km;
- Operation of ancillary buildings including E&M plant room to support the operation of depressed road and underpass;
- Permanent removal of existing trees and other vegetation due to the proposed road works and associated slope works; and
- Implementation of landscape and visual mitigation measures such as aesthetically pleasing design of aboveground structures, buffer screen planting and landscape treatments on slope or retaining structure etc. as detailed in **Table 10.9**.

Magnitude of Visual Impact on existing VPs

10.7.3 The magnitude of impacts during operational phase would be assessed and descripted subject to the visual composition, visual obstruction and visual change in text and shown in **Table 10.7** in accordance with the latest version of EIAO GN. Photomontages are prepared for the proposed VPs as per the requirements of the Study.

10.7.4 The criteria for the selection of representative VPs for photomontages include:-

- the VPs which cover the aboveground structure(s) viewed from major public VPs would be potentially affected by the proposed permanent structures; and
- the VPs should be able to represent the worst-case scenarios and demonstrate the compatibility of the aboveground structures to the adjacent visual context and illustrate the visual effect with mitigation measures.

Significance Thresholds of Visual Impact

10.7.5 The significance thresholds of visual impact, in the operational phase will be assessed and presented in **Table 10.11**.

Table 10.7 Magnitude of Visual Change during Operational Phase

VP ID.	Location of VPs	Source of Visual Impact	Visual Composition	Visual Obstruction	Visual Change	Magnitude of Change (Negligible / Slight / Moderate / Substantial)
VP1	View from Ma Tso Lung Fishpond	The proposed road works (underpass with ventilation opening) and associated structure	- The view is dominated by an open view of open sky and vast landscape of LMC. - The backdrop consists of the Shenzhen skyline and the Loop, while the fishponds in LMC occupy the middle ground with adjacent greenery and shrubs. - Most of the proposed road works and associated structure of the underpass are underground and not visible in this view. Only a small portion of the structure of ventilation opening for the proposed underpass would be visible and form part of the middle ground of this view.	- Subject to the medium viewing distance, low structural profile and minimal structural mass, the proposed development would have no obstruction to the existing visual elements. - The public viewers can still enjoy the existing panorama view of LMC, the Loop and the skyline of Shenzhen.	- The proposed road works and associated structure would be an additional visual element of this view. - The proposed road works and its associated structure would not be noticeable by the viewer due to minimal structural mass aboveground. - Effect of visual change on the public viewers: Negligible - Degree of visibility: Glimpse - Minimum viewing distance: 250m	Negligible

VP ID.	Location of VPs	Source of Visual Impact	Visual Composition	Visual Obstruction	Visual Change	Magnitude of Change (Negligible / Slight / Moderate / Substantial)
			- The visual compatibility of the proposed road works with surrounding context is considered as high.			
VP2	View from Ma Tso Lung Eco Park	The proposed road works (at grade and viaduct) and ancillary works, including slope works	- The view is dominated by a vista of open sky and natural hillside vegetation of Ma Tso Lung mountains. - The backdrop consists of the continuous ridgeline formed by Lei Yue Shan, Tai Shek Mo and nearby mountains, while village settlements, existing Border Road and adjacent greenery occupy the middle ground, and fishpond in the foreground. - The proposed road works and ancillary works, including	- Despite a small portion of existing view towards the natural hillside vegetation of Tai Shek Mo would be blocked by the proposed road works (at grade and viaduct), the visual openness of the existing visual elements, such as the open sky and continuous ridgeline, would be largely retained. - The overall visual permeability would be largely preserved due to the slender and airy structural design.	- The proposed road works and ancillary works would be an additional visual element at the middle ground of this view. - Since the proposed road works would cut though the natural hillside vegetation, the existing view of vast landscape would be alternated, and the greenery would be re-established with appropriate landscape treatment on the proposed slope works. - The visual change would be partially visible by the public viewers, with row of	Moderate

VP ID.	Location of VPs	Source of Visual Impact	Visual Composition	Visual Obstruction	Visual Change	Magnitude of Change (Negligible / Slight / Moderate / Substantial)
			slope works, are planned to be situated along the existing Border Road near Ma Tso Lung Eco Park. - The visual compatibility of the proposed road and slope works with surrounding context is considered as slight to moderate, with the fishpond at the foreground and open sky at the background maintained.		existing trees along the existing border road preserved as a green buffer to the proposed works. - Effect of visual change on the public viewers: Slight to moderate - Degree of visibility: Partial - Minimum viewing distance: 80m	
VP3	View from Macintosh Fort (Ma Tso Lung)	The proposed road works (at grade and viaduct)	- The elevated view is dominated by a panoramic view of open sky and vast landscape of Ma Tso Lung. - The backdrop consists of the continuous ridgeline formed by Ki Lun Shan and nearby	- Most of the proposed road works at the foothill is not visible in this view, it would have no obstruction to the existing visual elements. - The public viewers can still enjoy the existing panorama	- The value of this view is primarily attached to the visual amenity of panoramic view toward the open sky view and the continuous ridgeline in Ki Lun Shan, Kwu Tung and Ma Tso Lung.	Slight

VP ID.	Location of VPs	Source of Visual Impact	Visual Composition	Visual Obstruction	Visual Change	Magnitude of Change (Negligible / Slight / Moderate / Substantial)
			mountains in Kwu Tung, while natural hillside vegetation in She Leng and Ma Tso Lung wetland occupies the middle ground. - The presence of natural hillside vegetation in the foreground adds to the overall framing of the vista for the viewer. - The proposed road works are planned to be situated at the foothill of Lei Yue Shan and Tai Shek Mo. - The visual compatibility of the proposed works with the surrounding context is considered as high.	view of Ma Tso Lung and She Leng.	- Most of the proposed works would not be noticeable due to the alignment and blockage of existing visual elements. - The open sky view and vast landscape would still be the key visual elements of this VP. - Viewers at this VP are mainly hikers which are transient in nature. - Effect of visual change on the public viewers: Slight - Degree of visibility: Glimpse to partial - Minimum viewing distance: 170m	
VP4	View from footpath of	The proposed road works (at grade and	- The view is dominated by an open view of open	- Subject to the medium viewing distance and minimal	- The proposed works would be an	Moderate

VP ID.	Location of VPs	Source of Visual Impact	Visual Composition	Visual Obstruction	Visual Change	Magnitude of Change (Negligible / Slight / Moderate / Substantial)
	Ma Tso Lung San Tsuen	viaduct) and ancillary works, including slope works	sky and vast landscape of Ma Tso Lung, featuring the continuous ridgeline formed by Lei Yue Shan, Shun Yee San Tsuen and Ma Tso Lung San Tsuen at the background and the Ma Tso Lung wetland at the middle ground. - The presence of vegetation in the foreground adds to the overall framing of the vista for the viewer. - The proposed road works (at grade and viaduct) and ancillary works, including slope works, would form part of the visual backdrop of this view. - The visual compatibility of the proposed works with the surrounding	structural mass, the proposed development would have slight obstruction to the existing greenery. - The visual openness of the existing visual elements, such as the open sky and continuous ridgeline, would be largely retained. - The overall visual permeability would be largely preserved through slender viaduct design. - Part of the existing foothill right behind Shun Yee San Tsuen would be retained and as a green berm to partially screen the proposed retaining structure along the foothill of Lei Yue Shan.	additional visual element of this VP. - Despite the addition of the proposed road works and ancillary works, the open sky view and ridgeline of Lei Yue Shan would still be the key visual elements of this VP. - The proposed road works and ancillary works would be partially visible to viewers in medium viewing distance due to the blockage of existing visual elements. - Although the existing view of natural hillside would be alternated, the greenery would be re-established with appropriate landscape treatment on the proposed slope works and the	

VP ID.	Location of VPs	Source of Visual Impact	Visual Composition	Visual Obstruction	Visual Change	Magnitude of Change (Negligible / Slight / Moderate / Substantial)
			context is considered as moderate.		visual character of this VP would generally be retained. - Effect of visual change on the public viewers: Moderate - Degree of visibility: Partial - Minimum viewing distance: 300m	
VP5	View from She Leng towards Ma Tso Lung	The proposed road works (at grade and viaduct) and ancillary works, including slope works	- The elevated view is dominated by a panoramic view of open sky and vast landscape of Ma Tso Lung, featuring the continuous ridgeline formed by Lei Yue Shan and Tai Shek Mo, and Shun Yee San Tsuen at the background, with the Ma Tso Lung wetland and nearby natural hillside vegetation at the middle ground. - The presence of vegetation in the	- Subject to the long viewing distance and minimal structural mass, the proposed development would have slight obstruction to the existing greenery. - The visual openness of the existing visual elements, such as the open sky and continuous ridgeline, would be largely retained. - The overall visual permeability would be largely preserved due	- The proposed works would be an additional visual element of this VP. - Despite the addition of the proposed road works and ancillary works, the open sky view, ridgeline of Tai Shek Mo and nearby natural hillside vegetation would still be the key visual elements of this VP. - The proposed road works and ancillary works would be partially visible to	Moderate

VP ID.	Location of VPs	Source of Visual Impact	Visual Composition	Visual Obstruction	Visual Change	Magnitude of Change (Negligible / Slight / Moderate / Substantial)
			foreground adds to the overall framing of the vista for the viewer. - The proposed road works (at grade and viaduct) and ancillary works, including slope works, would form part of the visual backdrop of this view. - The visual compatibility of the proposed works with the surrounding context is considered as slight to moderate.	to the slender and airy structural design.	viewers due to long viewing distance and blockage of existing visual elements. - Although the existing view of natural hillside would be slightly alternated, the greenery would be re-established with appropriate landscape treatment on the proposed slope works and the visual character of this VP would generally be retained. - Viewers at this VP are mainly hikers which are transient in nature. - Effect of visual change on the public viewers: Slight to moderate - Degree of visibility: Partial	

VP ID.	Location of VPs	Source of Visual Impact	Visual Composition	Visual Obstruction	Visual Change	Magnitude of Change (Negligible / Slight / Moderate / Substantial)
					- Minimum viewing distance: 750m	
VP6	View from She Leng towards Hoo Hok Wai	The proposed road works (at grade)	- The elevated view is dominated by a panoramic view of open sky and vast landscape of Hoo Hok Wai, featuring the Shenzhen skyline at the background, the Loop and Hoo Hok Wai fishponds at the middle ground and framed by the green mountain slopes of She Leng at the foreground. - The proposed road works near She Leng are planned to be situated at the foothill, with similar alignment to the existing Border Road. - Only a tiny portion of the proposed road at the foothill is visible at this view, the visual compatibility of the	- Subject to the visibility and similar alignment and massing as compared to the existing Border Road at foothill of She Leng, the proposed development would have no obstruction to the existing visual elements. - The visual openness of the existing visual elements, such as the open sky and vast landscape of Hoo Hok Wai, would be largely retained. - The public viewers can still enjoy the existing panorama view of Hoo Hok Wai and skyline of Shenzhen.	- The value of this view is primarily attached to the open sky, vast landscape of Hoo Hok Wai and Shenzhen skyline. - The proposed road works and its associated slope works would not be noticeable by the viewer, due to the visibility and in similar nature and alignment to the existing Border Road. - The open sky view and vast landscape would still be the key visual elements of this VP. - Viewers at this VP are mainly hikers which are transient in nature.	Negligible

VP ID.	Location of VPs	Source of Visual Impact	Visual Composition	Visual Obstruction	Visual Change	Magnitude of Change (Negligible / Slight / Moderate / Substantial)
			proposed works with the surrounding context is considered as high.		- Effect of visual change on the public viewers: Negligible - Degree of visibility: Glimpse - Minimum viewing distance: 130m	

10.8 Landscape and Visual Mitigation Measures

The proposed mitigation measures presented in the approved Development of Lok Ma Chau Loop EIA Report (Register No.: AEIAR-176/2013) that is applicable to the Project are reviewed and referenced in this Report.

10.8.1 The proposed landscape and visual mitigation measures, together with an indication of funding, implementation and maintenance agencies, during the construction and operational phases are listed in **Table 10.8** and **Table 10.9** below. The mitigation measures proposed would be further reviewed and consents from relevant governmental department would be sought subject to the design development of the Project.

Table 10.8 Proposed Landscape Mitigation Measures for Construction Phase

ID No.	Landscape Mitigation Measures	Funding Agency	Implementation Agency
CM1	**Tree Preservation and Transplantation** Trees without impact from the proposed works should be retained and any existing trees unavoidably affected by the works should be transplanted as far as possible in accordance with DEVB TC(W) No. 4/2020.	CEDD	CEDD (via Contractor)
CM2	**Erection of Decorative Screen Hoarding** Erection of decorative screen hoarding or hoarding compatible with the surrounding setting.	CEDD	CEDD (via Contractor)
CM3	**Management of Construction Activities and Facilities** Construction facilities and activities on work sites and areas should be carefully managed and controlled, including height and disposition /arrangement, to minimise any potential adverse landscape impacts.	CEDD	CEDD (via Contractor)
CM4	**Reinstatement of the Affected Landscaped Area** Reinstatement of the temporarily affected landscaped area within assessment area in like-for-like basis would be implemented to restore the existing natural environment as far as possible.	CEDD	CEDD (via Contractor)
CM5	**Coordination with Concurrent Projects** Coordinated implementation programme with concurrent projects to minimise impacts and where possible reduce the period of disturbance.	CEDD	CEDD (via Contractor)

Note: “CM” stands for mitigation measure for construction phase.

Table 10.9 Proposed Landscape and Visual Mitigation Measures for Operational Phase

ID No.	Landscape and Visual Mitigation Measures	Funding Agency	Implementation Agency	Maintenance/ Management Agency*
OM1	**Aesthetically Pleasing Design of Aboveground Structures** Aesthetically pleasing design as regard to the form, material and finishes should be incorporated into all aboveground structures, roads, viaducts, ancillary buildings and other associated engineering facilities so as to blend in the structures with the adjacent landscape and visual context.	CEDD	CEDD (via Contractor)	HyD
OM2	**Buffer Screen Planting** Buffer screen planting, including shrubs, to provide screening.	CEDD	CEDD (via Contractor)	HyD / LCSD
OM3	**Compensatory Tree Planting** As far as practicable, compensatory tree planting requirement of 1:1 in terms of number in accordance with DEVB TC(W) No. 4/2020 for compensation of felled trees subject to the availability of space and agreement made with relevant departmental parties.	CEDD	CEDD (via Contractor)	HyD / LCSD
OM4	**Reinstatement of Affected Fishponds** Temporary loss of fishponds along ECR by the road widening and improvement works will be largely reinstated. These ponds will be used for both functional or amenity purposes to enhance the existing landscape and visual context.	CEDD	CEDD (via Contractor)	HyD / AFCD
OM5	**Landscape Treatments on Slope or Retaining Structure** Landscape treatments on slope or retaining structure in accordance with GEO Publication No. 1/2011 – Technical Guidelines on Landscape Treatment for Slopes shall be adopted subject to the availability of space and agreement made with relevant departmental parties.	CEDD	CEDD (via Contractor)	HyD

Note: "OM" stands for mitigation measure for operational phase. Management and Maintenance Agencies are identified as per DEVB TC(W) No. 6/2015.

10.8.2 The construction phase mitigation measures listed above should be adopted from the commencement of construction and should be in place throughout the entire construction period. The operational phase mitigation measures listed above should be adopted during the detailed design and should be built as part of the construction

works so that they are in place at the date of commissioning of the Project. However, it should be noted that the full effect of the soft landscape mitigation measures would not be appreciated for several years.

10.8.3 Viewpoint locations of the photomontages are shown in **60722948/A10/FIGURE 10.32**. The photomontages intend to demonstrate only the scale and massing of the Project and the effect of the proposed mitigation measures. The architectural design finishes or any other related detailed design components are subject to refinement and changes during the detailed design stage. Photomontages of the Project without and with mitigation measures illustrating the appearance of the Project are shown in **60722948/A10/FIGURE 10.33** to **10.38**.

10.9 Landscape and Visual Impacts

10.9.1 After implementation of recommended mitigation measures shown in the Landscape and Visual Mitigation Plans in **60722948/A10/FIGURE 10.39** to **10.45**, Landscape and Visual Mitigation Sections in **60722948/A10/FIGURE 10.46** to **10.48**, residual landscape impact of significance during construction and operational phases are provided in **Table 10.10** and described below.

Landscape Resources

10.9.2 For LR1, LR2, LR5 and LR8, there would be no residual impact on these LRs during construction and operation as no construction works would be conducted in these LRs.

LR 3 – Fishponds

10.9.3 During the construction phase, a small portion of water body would be taken up for the proposed works. The anticipated unmitigated landscape impact would be **Moderate** due to permanent loss of natural resources. Upon the completion of works, provision of buffer planting and reinstatement of affected fishponds would be proposed to alleviate the landscape impact to this LR. It is considered that residual impact on this LR would be **Slight** during the construction phase and in operation with the implementation of mitigation measures.

LR 4 – Marsh / Wetland

10.9.4 During the construction phase, a small portion of existing marsh / wetland and adjacent vegetation would be permanently affected by the proposed works. The anticipated unmitigated landscape impact would be **Moderate** due to permanent loss of natural resources. Upon the completion of works, provision of buffer planting and tree compensation would be proposed to alleviate the landscape impact to this LR. It is considered that residual impact on this LR would be **Slight** during the construction phase and in operation with the implementation of mitigation measures.

LR 6 – Mixed Woodland

10.9.5 During the construction phase, a small portion of existing mixed woodland vegetation would be permanently affected by the proposed works, the unmitigated landscape impact on this LR would be considered **Moderate** due to permanent loss of natural resources. After the completion of works, the natural vegetated slope would be partially replaced as man-made slope with woodland mix slope planting for greening. The affected hillside vegetation would be re-established over the time. It is considered that residual impact on this LR would be **Moderate** during the construction phase and

reduced to **Slight** in operation with the implementation of mitigation measures after establishment of the compensatory planting.

LR 7 – Grassland / Shrubland

10.9.6 During the construction phase, some portion of existing grassland / shrubland vegetation would be permanently affected by the proposed works, the unmitigated landscape impact on this LR would be considered **Moderate** due to permanent loss of natural resources. After the completion of works, the natural vegetated slope would be partially replaced as man-made slope with woodland mix slope planting for greening. The affected vegetation would be re-established over the time. It is considered that residual impact on this LR would be **Slight** during the construction phase and in operation with the implementation of mitigation measures.

LR 9 – Vegetation in Village Settlement

10.9.7 During the construction phase, some portion of vegetation identified within the village settlement would be permanently affected by the provision of elevated infrastructure. The anticipated unmitigated landscape impact would be **Slight**. Upon the completion of works, provision of buffer planting and tree compensation would be proposed to alleviate the landscape impact to this LR. It is considered that residual impact on this LR would be **Slight** during the construction phase and reduced to **Negligible** in operation with the implementation of mitigation measures after establishment of the compensatory planting.

LR 10 – Vegetation in Developed Area and Roads

10.9.8 Small cluster of man-made vegetation identified within the developed area would be permanently affected due to the proposed works. The anticipated unmitigated landscape impact would be **Slight**. Upon the completion of works, provision of greenery would be subject to the future development in the developed area. It is considered that residual impact on this LR would be **Negligible** during the construction phase and in operation with the implementation of mitigation measures.

LR 11 – Roadside Planting

10.9.9 During the construction phase, some portion of engineered slope would be affected by the proposed works, the unmitigated landscape impact on this LR would be considered **Slight**. Upon the completion of works, the temporarily affected engineered slope would be reinstated with woodland mix slope planting while some of the engineered slope would be permanently occupied by the proposed road works. The affected vegetation would be re-established over the time. It is considered that residual impact on this LR would be **Negligible** during the construction phase and in operation with the implementation of mitigation measures.

Landscape Character Areas

LCA 1 – Miscellaneous Urban Fringe Landscape of the Loop

10.9.10 Given to the removal or relocation of small portion of buildings in the developed area, it is anticipated that the unmitigated landscape impact would be **Slight**. Since the temporarily affected buildings or facilities would be redeveloped by others, the character of this LCA would still largely be preserved. Hence, it is assumed that that residual impact this LCA would be reduced to **Negligible** during the construction phase and in operation with the implementation of mitigation measures.

LCA 2 – Lowland Rural Landscape of Lok Ma Chau

10.9.11 Upon the completion of works, only a small portion of the periphery of the existing lowland landscape would be occupied by the proposed works. The unmitigated landscape impact would be **Slight**. Landscape mitigation measures such as roadside greenery, buffer planting at engineered slopes as compensation as compensation would be adopted to mimic the current site condition and alleviate the potential landscape impact. The affected vegetated area would be re-established throughout time. Hence, it is considered that the residual impact on this LCA would be reduced to **Slight** during the construction phase and reduced to **Negligible** in operation with the implementation of mitigation measures after establishment of the compensatory planting.

LCA 3 – Hillside Landscape of Lok Ma Chau

10.9.12 Upon the completion of works, only a small portion of the periphery of the hillside vegetation would be occupied by the proposed works. The anticipated unmitigated landscape impact would be **Moderate** due to permanent loss of natural resources with high sensitivity. Landscape mitigation measures such as reinstatement of affected landscaping area, woodland mix planting at engineered slopes as compensation would be adopted to alleviate the potential landscape impact. Vegetation would be further re-established throughout time to re-create the current site character. Hence, it is anticipated that the residual impact on this LCA would be reduced to **Slight** during the construction phase and in operation with the implementation of mitigation measures.

LCA 4 – Lowland Rural Landscape of Hoo Hok Wai

10.9.13 Upon the completion of works, only a small portion of the periphery of the existing lowland landscape would be occupied by the proposed works. The anticipated unmitigated landscape impact would be **Moderate** due to permanent loss of natural resources with high sensitivity. Landscape mitigation measures such as roadside greenery, buffer planting at engineered slopes as compensation would be adopted to mimic the current site condition and alleviate the potential landscape impact. The affected vegetated area would be re-established throughout time. Hence, it is considered that the residual impact on this LCA would be reduced to **Slight** during the construction phase and in operation with the implementation of mitigation measures.

LCA 5 – Lowland Rural Landscape of Ma Tso Lung

10.9.14 Upon the completion of works, a small portion of the existing lowland landscape would be occupied by the proposed works. The anticipated unmitigated landscape impact would be **Slight**. Landscape mitigation measures such as roadside greenery, buffer planting at engineered slopes as compensation would be adopted to mimic the current site condition and alleviate the potential landscape impact. The affected vegetated area would be re-established throughout time. Hence, it is considered that the residual impact on this LCA would be **Slight** during the construction phase and reduced to **Negligible** in operation with the implementation of mitigation measures after establishment of the compensatory planting.

LCA 6 – Hillside Landscape of Ma Tso Lung

10.9.15 Given that the alignment of the proposed works would cut though the vegetated hillside of Ma Tso Lung, a portion of existing vegetated hillside would be permanently

occupied by the highway structures while the natural hillside with slope cutting works resulted in the provision of man-made engineered slope. The anticipated unmitigated landscape impact would be **Moderate** due to permanent loss of natural resources with high sensitivity. Landscape mitigation measures such as reinstatement of affected landscaping area, woodland mix planting at engineered slopes as compensation would be adopted to alleviate the potential landscape impact. Vegetation would be further re-established throughout time to re-create the current site character. Hence, it is anticipated that the residual impact on this LCA would be **Moderate** during the construction phase and reduced to **Slight** in operation with the implementation of mitigation measures after establishment of the compensatory planting.

Table 10.10 Significance Thresholds of Landscape Impact during Construction and Operational Phases

ID No.	Landscape Resource / Landscape Character Areas	Sensitivity (Low, Medium, High)		Magnitude of Change (Negligible, Slight, Moderate, Substantial)		Impact Significance Thresholds before Mitigation (Negligible, Slight, Moderate, Substantial)		Recommended Mitigation Measures (1)	Significance Thresholds of Residual Impact after Mitigation (Beneficial, Negligible, Slight, Moderate, Substantial)	
		Construction	Operation	Construction	Operation	Construction	Operation		Construction	Operation
Landscape Resources (LRs)										
LR1	Natural Stream / River Course	High	High	Negligible	Negligible	Negligible	Negligible	-	Negligible	Negligible
LR2	Engineered Water Channel	Low	Low	Negligible	Negligible	Negligible	Negligible	-	Negligible	Negligible
LR3	Fishponds	High	High	Slight	Slight	Moderate	Moderate	CM2, CM3, CM4, OM1, OM4	Slight	Slight
LR4	Marsh / Wetland	High	High	Slight	Slight	Moderate	Moderate	CM1, CM2, CM3, CM4, OM1, OM2, OM3, OM5	Slight	Slight
LR5	Man-made Marsh / Wetland	Medium	Medium	Negligible	Negligible	Negligible	Negligible	-	Negligible	Negligible
LR6	Mixed Woodland	High	High	Moderate	Moderate	Moderate	Moderate	CM1, CM2, CM3, CM4, OM1, OM2, OM3, OM5	Moderate	Slight
LR7	Grassland / Shrubland	Medium	Medium	Moderate	Moderate	Moderate	Moderate	CM1, CM2, CM3, CM4, OM1, OM2, OM3, OM5	Slight	Slight
LR8	Agricultural Fields	Medium	Medium	Negligible	Negligible	Negligible	Negligible	-	Negligible	Negligible

ID No.	Landscape Resource / Landscape Character Areas	Sensitivity (Low, Medium, High)		Magnitude of Change (Negligible, Slight, Moderate, Substantial)		Impact Significance Thresholds before Mitigation (Negligible, Slight, Moderate, Substantial)		Recommended Mitigation Measures (1)	Significance Thresholds of Residual Impact after Mitigation (Beneficial, Negligible, Slight, Moderate, Substantial)	
		Construction	Operation	Construction	Operation	Construction	Operation		Construction	Operation
LR9	Vegetation in Village Settlement	Medium	Medium	Slight	Slight	Slight	Slight	CM1, CM2, CM3, CM4, CM5, OM1	Slight	Negligible
LR10	Vegetation in Developed Area	Low	Low	Slight	Slight	Slight	Slight	CM1, CM2, CM3, CM4, CM5, OM1, OM2, OM3	Negligible	Negligible
LR11	Roadside Planting	Low	Low	Slight	Slight	Slight	Slight	CM1, CM2, CM3, CM4, OM1, OM2, OM3, OM5	Negligible	Negligible
Landscape Character Areas (LCAs)										
LCA 1	Miscellaneous Urban Fringe Landscape of the Loop	Low	Low	Slight	Slight	Slight	Slight	CM1, CM2, CM3, CM4, CM5, OM1, OM2	Negligible	Negligible
LCA 2	Lowland Rural Landscape of Lok Ma Chau	Medium	Medium	Slight	Slight	Slight	Slight	CM1, CM2, CM3, CM4, OM1, OM2, OM4	Slight	Negligible
LCA 3	Hillside Landscape of Lok Ma Chau	High	High	Slight	Slight	Moderate	Moderate	CM1, CM2, CM3, CM4, OM1, OM2, OM3, OM5	Slight	Slight
LCA 4	Lowland Rural Landscape of Hoo Hok Wai	High	High	Slight	Slight	Moderate	Moderate	CM1, CM2, CM3, CM4, OM1, OM2, OM3, OM5	Slight	Slight

ID No.	Landscape Resource / Landscape Character Areas	Sensitivity (Low, Medium, High)		Magnitude of Change (Negligible, Slight, Moderate, Substantial)		Impact Significance Thresholds before Mitigation (Negligible, Slight, Moderate, Substantial)		Recommended Mitigation Measures (1)	Significance Thresholds of Residual Impact after Mitigation (Beneficial, Negligible, Slight, Moderate, Substantial)	
		Construction	Operation	Construction	Operation	Construction	Operation		Construction	Operation
LCA 5	Lowland Rural Landscape of Ma Tso Lung	Medium	Medium	Slight	Slight	Slight	Slight	CM1, CM2, CM3, CM4, CM5, OM1, OM2, OM3, OM5	Slight	Negligible
LCA 6	Hillside Landscape of Ma Tso Lung	High	High	Moderate	Moderate	Moderate	Moderate	CM1, CM2, CM3, CM4, OM1, OM2, OM3, OM5	Moderate	Slight

Note:
(1) Refer to **Table 10.8** - Proposed Landscape Mitigation Measures for Construction Phase & **Table 10.9** - Proposed Landscape and Visual Mitigation Measures for Operational Phase

Visual Impact to existing VPs

10.9.16 With the implementation of the recommended mitigation measures shown in the Landscape and Visual Mitigation Plans in Figures **60722948/A10/FIGURE 10.39** to **10.45**, Landscape and Visual Mitigation Sections in **60722948/A10/FIGURE 10.46** to **10.48**, significance thresholds of visual impact during operational phase are described below. Photomontages illustrating the potential visual impact during operation with and without mitigation measures from the VPs are shown in **60722948/A10/FIGURE 10.33** to **10.38**.

VP1 View from Ma Tso Lung Fishpond

10.9.17 Given the minimal structural mass of the proposed works aboveground, the anticipated visual change would not be noticeable by viewers due to medium viewing distance. The magnitude of change to this VP would be considered as **Negligible**. Hence, it is expected that visual impact after implementation of mitigation measure would be **Negligible** in operational phase.

VP2 View from Ma Tso Lung Eco Park

10.9.18 The proposed works would occupy the middle ground of this VP in short viewing distance. After the implementation of mitigation measures such as aesthetic pleasing treatment of viaduct, landscape treatment on slope and buffer screening planting, the visual changes would be slight to moderate with row of existing trees along the existing border road preserved as a green buffer to the proposed works. The magnitude of change to this VP would be considered as **Moderate** with medium sensitivity. It is anticipated that the visual impact for the viewers of this VP would be **Moderate** in operational phase.

VP3 View from Macintosh Fort (Ma Tso Lung)

10.9.19 Most of the proposed road works at the foothill is not visible in this view, most of the anticipated visual change would not be noticeable by viewers due to the road alignment and blockage of existing visual elements. The magnitude of change to this VP would be considered as **Slight**. Hence, it is expected that visual impact after implementation of mitigation measure would be **Slight** in operational phase.

VP4 View from footpath of Ma Tso Lung San Tsuen

10.9.20 Given to the medium viewing distance, the proposed works would occupy the middle ground of this VP, existing view of natural hillside would be alternated while the open sky view and ridgeline remain unchanged. To alleviate the visual changes caused by the proposed works, several mitigation measures would be proposed, such as provision of appropriate landscape treatment on slope and aesthetically pleasing design of aboveground structure. After the implementation of the mitigation measures, the magnitude of change to this VP would be considered as **Moderate.** It is anticipated that the visual impact for the viewers of this VP with low sensitivity would be **Slight** in operational phase.

VP5 View from She Leng towards Ma Tso Lung

10.9.21 Despite the addition of the proposed works, the open sky view and ridgeline of Tai Shek Mo and nearby natural hillside vegetation would still be the key visual elements of this VP and would be largely preserved. To alleviate the visual changes caused by

the proposed works, several mitigation measures would be proposed, such as provision of appropriate landscape treatment on slope and aesthetically pleasing design of aboveground structure. The visual changes would be slight to moderate and the magnitude of change to this VP would be considered as **Moderate**. After the implementation of the mitigation measures, it is anticipated that the visual impact for the viewers of this VP would be **Moderate** in operational phase given the high sensitivity of this VP.

VP6 View from She Leng towards Hoo Hok Wai

10.9.22 Most of the proposed road works at the foothill is not visible in this view as blockage by the existing topography and vegetation. The anticipated visual change would not be noticeable by viewers due to the visibility and in similar nature and alignment to the existing Border Road. The magnitude of change to this VP would be considered as **Negligible**. Hence, it is expected that visual impact after implementation of mitigation measure would be **Negligible** in operational phase.

Table 10.11 Significance Thresholds of Visual Impact during Operational Phase

VP ID.	Location of VPs / Key Public Viewers	Sensitivity (High, Medium, Low)	Magnitude of Change (Negligible, Slight, Moderate, Substantial)	Recommended Mitigation Measures (2)	Significance Thresholds of Visual Impact after Mitigation Measure Established (Beneficial, Negligible, Slight, Moderate, Substantial)
VP1	View from Ma Tso Lung Fishpond	Medium	Negligible	OM1, OM4	Negligible
VP2	View from Ma Tso Lung Eco Park	Medium	Moderate	OM1, OM2, OM3, OM5	Moderate
VP3	View from Macintosh Fort (Ma Tso Lung)	Medium	Slight	OM1, OM3, OM5	Slight
VP4	View from footpath of Ma Tso Lung San Tsuen	Low	Moderate	OM1, OM3, OM5	Slight
VP5	View from She Leng towards Ma Tso Lung	High	Moderate	OM1, OM3, OM5	Moderate
VP6	View from She Leng towards Hoo Hok Wai	High	Negligible	OM1, OM3, OM5	Negligible

Note:
(2) Refer to **Table 10.9** - Proposed Landscape and Visual Mitigation Measures for Operational Phase

Evaluation of the Residual Landscape Impacts and Visual Impacts

10.9.23 Considering the scale and nature of the Project, it would inevitably result in certain levels of residual landscape impacts and visual impacts in relation to the loss of hillside vegetation (i.e. LR6, LR7, LCA3 and LCA6), loss of lowland vegetation (i.e. LR3, LR4, LR9 and LCA4) and the views from hilltop and from ground level (i.e. VP2, VP3, VP4, and VP5). The residual landscape impacts and visual impacts are evaluated below in accordance with Section 4.4.3 of EIAO-TM:

- The magnitude of the adverse environmental impacts:

  The magnitude of residual landscape impacts of LR3, LR4, LR6, LR7, LCA3, LCA4 and LCA6 are slight, and the magnitude of visual impacts of VP2, VP3, VP4, and VP5 are slight to moderate, and minor in scale. The proposed development incorporating landscape mitigation measures such as tree compensation where space is available, reinstatement of affected landscaping area or fishpond in like-for-like, provision of buffer screen planting and appropriate landscape treatment on slope and retaining structures would be implemented. Visually compatible design had also been developed for the proposed ECR with slender and airy structural design, to ameliorate the overall landscape and visual qualities.

- The geographic extent of the adverse environmental impacts:

  The residual landscape impacts and visual impacts would not cause adverse environmental impact on the geographical extent. For LR6, LR7, LCA3 and LCA6, loss of hillside vegetation is localised and minimised, and no country park would be affected. For VP2, VP3, VP4, and VP5, all the visual impacts are confined within the visual envelope.

- The duration and frequency of the adverse environmental impacts:

  The residual landscape impacts and visual impacts are long term and permanent due to the nature of road construction and associated works, transforming hillside vegetation to man-made slope. The durations of view on VP2 and VP4 are short, as most viewers along the path are transient. The durations of view are short to medium for VP3 and VP5, as most of the public viewers may take short rest to enjoy the scenery and then continue their hiking journey.

- The likely size of the community or the environment that may be affected by the adverse impacts:

  The residual landscape impacts are localised and confined to the loss of hillside vegetation. No existing community in Ma Tso Lung would be involved, and no country park would be affected. The visual impacts on VP2, VP3, VP4 and VP5 involved relatively few to medium number of public viewers, mainly hikers in She Leng and to Macintosh Fort (Ma Tso Lung), and recreational users / travellers in Ma Tso Lung Eco Park.

- The degree to which the adverse environmental impacts are reversible:

  The residual landscape impacts are irreversible due to the nature of road construction and associated slope works, transforming hillside vegetation to

man-made slope. The visual impacts are also long term and irreversible with the introduction of new road in the area. Nevertheless, the proposed development incorporating landscape mitigation measures such as tree compensation where space is available, reinstatement of affected landscaping area or fishpond in like-for-like basis, provision of buffer screen planting and appropriate landscape treatment on slope and retaining structures would be implemented. Visually compatible design had also been developed for the proposed ECR, with slender and airy structural design, to ameliorate the overall landscape and visual qualities.

- International and regional importance:

  The residual landscape impacts and visual impacts would not involve any distinctive landscape resource, landscape character and vantage viewing point with international or regional importance.

- Both the likelihood and degree of uncertainly of adverse environmental impacts:

  The residual landscape impacts and visual impacts would be foreseeable without uncertainty at the time of EIA preparation.

10.9.24 As discussed above, the residual landscape impacts are slight and visual impacts ranged from slight to moderate, long term and irreversible due to the nature of road construction and associated slope works, transforming hillside vegetation to man-made slope, are inevitable. Nevertheless, the residual landscape impacts are localised and limited to the road construction and associated works extent only without affecting existing community, while the visual impacts are confined within the visual envelope either involving few numbers of public viewers along footpath or hiking trail, with only partially visible to viewers in medium to long viewing distance due to the blockage of existing visual elements. Although not all landscape and visual impacts can be fully reduced or eliminated through the implementation of mitigation measures. The design, construction and operation of the ECR would also fully comply with relevant ordinances, regulations, standards and guidelines as stipulated in **Section 10.3**. Hence, with full implementation of the recommended mitigation measures, unacceptable adverse residual landscape impacts and visual impacts are not expected, as evaluated based on the relevant factors in Section 4.4.3 of the EIAO-TM.

10.10 Cumulative Impacts

10.10.1 Cumulative landscape impacts during construction and operational phases from other key concurrent projects in the vicinity were assessed. The planned construction period and a brief description of the projects identified has been summarised in **Section 2.10** of the EIA report.

10.10.2 According to the latest implementation programme in **Section 2.9** of the EIA report, the construction works of the Project will tentatively commence in 2027 and completed by Q4 2031. A number of committed and planned developments that would likely interface with the implementation of Eastern Connection Road (ECR) were identified within the 100m assessment area of the Project. These include:-

- Development of the Loop

- Ma Tso Lung Area and Other Sites in Kwu Tung North New Development Area and North District – Zone 1 Ma Tso Lung Area

- Northern Metropolis Highway – Kwu Tung Section

10.10.3 "Development of the Loop" and "Ma Tso Lung Area and Other Sites in Kwu Tung North New Development Area and North District – Zone 1 Ma Tso Lung Area" would be respectively constructed and connected to the ECR. The construction works of "Development of Lok Ma Chau Loop" is commenced in 2021 for expected completion in phases in around 6 years, while the construction works of the "Ma Tso Lung Area and Other Sites in Kwu Tung North New Development Area and North District – Zone 1 Ma Tso Lung Area" is tentatively scheduled to commence in 2028 and the occupation / population intake in 2034. The "Northern Metropolis Highway – Kwu Tung Section" is expected to commission in about 3 to 4 years after the commissioning of the San Tin Section, in or before 2036.

10.10.4 The planned construction of "Development of the Loop" would potentially affect the existing vegetation at the Developed Area in the Loop (i.e. LR10), resulting in cumulative impact on the landscape resource during the early stage of construction of the Project. It is anticipated that there will not be any significant cumulative landscape impacts during the construction phase after implementation of mitigation measures such as management of construction activities and facilities and coordination with concurrent projects.

10.10.5 The Project would be completed around Q4 2031, while the planned construction of "Ma Tso Lung Area and Other Sites in Kwu Tung North New Development Area and North District – Zone 1 Ma Tso Lung Area" is tentatively scheduled to commence in 2028 and the occupation / population intake in 2034. Based on the available information at the time of this EIA study, it is expected that a small portion of the existing hillside vegetation near the east of Tai Shek Mo and existing vegetation in village settlements in Ma Tso Lung maybe affected resulting in cumulative impact on the landscape resources such as LR7 and LR9 during the middle to late stages of construction. Since the Project is expected to be completed around similar time as this planned development, it is anticipated that there will not be any significant cumulative landscape impacts during construction and operational phases after the implementation of mitigation measures such as buffer screen planting, compensatory tree planting and landscape treatments on slope or retaining structures.

10.10.6 The planned construction of "Northern Metropolis Highway – Kwu Tung Section" located at the eastern end of the assessment area, no cumulative landscape impact during construction phase is expected based on the available information at the time of this EIA study. Since the Project would be completed around Q4 2031, while the "Northern Metropolis Highway – Kwu Tung Section" in similar nature of road projects would be completed later in or before 2036, it is anticipated that there will not be any significant cumulative landscape impacts during the operational phase.

10.11 Environmental Monitoring and Audit

10.11.1 The detailed landscape and engineering design of the Project shall be undertaken so as to ensure compliance with the landscape and visual mitigation measures described in **Section 10.8**.

10.11.2 Implementation of the recommended mitigation measures would be regularly audited during construction phase and the 12-month establishment period during operational

phase. Details of environmental monitoring and audit requirement are stated in **Section 13.9** of the EIA Report.

10.12 Conclusion

10.12.1 The Project will inevitably result in some landscape and visual impacts during construction and operational phases. These impacts were minimised through the consideration on minimisation of works areas, incorporation of aesthetic external designs and landscape treatments in the proposed infrastructures.

10.12.2 Due to the proposed works, a portion of vegetation areas would be unavoidably affected. Based on the findings of broad brush tree and vegetation survey, approximately 2,642 nos. of existing trees identified within the 100m assessment area, in which approx. 2,315 nos. of existing trees would not be affected by the proposed works and would be retained. Among the approx. 1,019 nos. of existing trees within the Project Site Boundary, approx. 32% of existing trees (i.e. approx. 327 nos. of trees, including 4 nos. of TPIs (i.e. *Aquilaria sinensis*) and approx. 5 nos. of undesirable species (i.e. *Leucaena leucocephala))* would be unavoidably affected by the proposed works. Transplantation of all 4 nos. of affected TPIs (*Aquilaria sinensis* (土沉香)) is recommended. Tree compensation within the site in a form of woodland mix tree planting on slopes or heavy standard tree planting in amenity area would be explored. Tree compensation requirement of 1:1 in terms of number should be carried out as far as practicable. Exact number of trees to be retained, transplanted and removed to be determined under the Tree Preservation and Removal Proposals (TPRPs) in accordance with DEVB TC(W) – No. 4/2020.

10.12.3 Under this Project, the existing landscape resources, including vegetation, marsh or wetland and fishpond would be affected in multiple scale subject to the proposed works. Vegetation found within the LRs and LCAs would either be temporarily or permanently loss. After the completion of works, landscape mitigation measures such as tree compensation where space is available, reinstatement of affected landscaping area or fishpond in like for like basis and provision of buffer screen planting would be implemented. With the implementation of these mitigation measures, it is considered that residual impacts of the LRs and LCAs would be reduced to **Negligible** to **Slight** in the operational phase.

10.12.4 In terms of the landscape impact, a number of LRs would be preserved in total (i.e. LR1, LR2, LR5 and LR8). Among the remaining LRs, fishponds (LR3), marsh / wetland (LR4), mixed woodland (LR6) and grassland / shrubland (LR7) would have moderate impact significance before mitigation. Nevertheless, after the completion of works, landscape mitigation measures such as tree compensation where space is available, reinstatement of affected landscaping area or fishpond in like-for-like basis and provision of buffer screen planting would be implemented as appropriate. With the implementation of these mitigation measures, it is considered that residual impacts on these LRs would be reduced to **Slight** in operational phase.

10.12.5 Meanwhile, vegetation in village settlements (LR9), vegetation in developed area and roads (LR10) and roadside planting (LR11) would have slight impact significance before mitigation due to the proposed development. It is anticipated that residual impacts on these LRs would be reduced to **Negligible** in operational phase after the implementation of mitigation measures.

10.12.6 For LCAs, the most permanent works such as construction at-grade carriageway, associated structure and ancillary work, e.g. slope works, would be located within

hillside landscape of Lok Ma Chau (LCA3), lowland rural landscape of Hoo Hok Wai (LCA4) and hillside landscape of Ma Tso Lung (LCA6). Hence, it is anticipated that the impact significance before mitigation would be moderate due to their high sensitivity. With the implementation of mitigation measures, the residual impact of these LCAs would be reduced to **Slight** in operational phase.

10.12.7 The proposed works within lowland rural landscape of Lok Ma Chau (LCA2) and lowland rural landscape of Ma Tso Lung (LCA5) would slightly alter the existing landscape character. It is assumed that there would be slight impact significance before mitigation to these LCAs. With the implementation of mitigation measures, the residual impact would be reduced to **Negligible** in operational phase. For miscellaneous urban fringe landscape of the Loop (LCA1), the existing landscape character is similar to the proposed works, and it is assumed that there would be slight impact significance before mitigation. Hence, the residual impacts on this LCA would be reduced to **Negligible** in both construction phase and operational phase after the implementation of mitigation measures.

10.12.8 In terms of the visual impact, considered that the proposed works would have similar alignment to the existing infrastructures (i.e. Border Road) and minimal structural mass, the visual impact to the identified key public viewers would be ranging from **Negligible** to **Moderate**.

10.12.9 For VPs (i.e. VP1 and VP6) that viewing to LMC and Hoo Hok Wai, the proposed works are relatively minimal and with no significant visual blockages to existing visual elements. The anticipated visual impact of operational phase would be **Negligible**.

10.12.10 For VP (i.e. VP3) that viewing to She Leng and adjacent natural hillside landscape, as most of the proposed works are located at the foothill of Tai Shek Mo with blockage of existing visual elements, visual change would not be noticeable by viewers. The anticipated visual impact of operational phase would be **Slight**.

10.12.11 For VPs (i.e. VP2, VP4 and VP5) that viewing to Lei Yue Shan and adjacent natural hillside landscape, despite the proposed works would consider as new visual elements to the existing view, no significant visual blockages to existing key visual elements such as open sky and ridgeline of Lei Yue Shan. With the mitigation measures such as provision of greenery on slope and roadside area and aesthetically pleasing design of road and associated structures, the anticipated visual impact of operational phase for VP4 would be **Slight**, while the anticipated visual impacts operational phase for VP2 and VP5 would be **Moderate** in view of the medium to high sensitivity of these VPs.

10.12.12 Considering the scale and nature of the Project, it would inevitably result in certain levels of residual landscape impacts and visual impacts in relation to the loss of hillside vegetation from hilltop and from ground level. Nevertheless, the residual landscape impacts are localised and limited to the project extent only without affecting existing community, while the visual impacts are confined within the visual envelope either involving few to medium numbers of public viewers along hiking trail and paths. With the implementation of the proposed landscape and visual mitigation measures, the overall landscape residual impacts would be from **Negligible** to **Slight** in operational phase, and the overall visual impacts would be from **Negligible** to **Moderate** in operational phase. With full implementation of the recommended mitigation measures, unacceptable adverse residual landscape impacts and visual impacts are not expected.